Computers & Security 173 (2027) 105217

Contents lists available at ScienceDirect

Computers & Security

journal homepage: www.elsevier.com/locate/cose

Intrusion detection datasets for Cyber–Physical Systems: Taxonomy, challenges, and opportunities

Vagner E. Quincozes [a],*, Silvio E. Quincozes [b],*, Célio Albuquerque [a], Diego Passos [c], Daniel Mossé [d],*

[a] Computer Science Department, Universidade Federal Fluminense (UFF), Niterói/RJ, Brazil
[b] Campus Alegrete, Universidade Federal do Pampa (UNIPAMPA), Alegrete/RS, Brazil
[c] Instituto Superior de Engenharia de Lisboa (ISEL), Instituto Politécnico de Lisboa, Lisbon, Portugal
[d] Computer Science Department, University of Pittsburgh (PITT), Pittsburgh/PA, USA

ARTICLE INFO

Keywords:
Cyber–Physical Systems
Machine learning
Intrusion detection
Datasets
Dataset taxonomy

ABSTRACT

Cyber–Physical Systems (CPSs) enable several critical applications that need to be continuously hardened. However, in practice, they often rely on insecure standards (*e.g.*, IEC-61850 (Premaratne et al., 2010)) and protocols that prioritize timely delivery over secure communication. Thus, employing Intrusion Detection Systems (IDSs) to detect malicious activities is vital. Nevertheless, it is challenging for IDSs to accurately detect CPS cyberattacks since proper domain-specific datasets are scarce. In this work, we propose a taxonomy, review state-of-the-art CPS intrusion datasets with respect to how they provide features that enable intrusion detection, and discuss their challenges and opportunities. Our analysis includes an examination of the sources, types of attacks, and features of these datasets, as well as an in-depth analysis of their number of samples, attacks, and features. We also investigate the physical characteristics of the systems used to extract features from these datasets. Finally, we conduct experiments to validate the taxonomy's effectiveness in systematically identifying dataset limitations, highlighting gaps in attack coverage, feature representation, and data diversity, which are critical for IDS assessments.

Contents

1. Introduction 2
2. Taxonomy and related work 2
3. CPS intrusion detection dataset challenges 4
   3.1. The need for data 4
   3.2. Dataset suitability 5
   3.3. Data generation and collection 5
4. Intrusion detection datasets 5
   4.1. Applications and attacks 5
   4.2. Source and features 8
   4.3. Physical components 9
   4.4. Opportunities 9
5. Experiments 11
   5.1. Methodology and evaluation framework 11
   5.2. Results and discussion 12
   5.3. Feature contribution analysis 14
6. Conclusion 16
   CRediT authorship contribution statement 16
   Declaration of competing interest 16

* Corresponding authors.
E-mail addresses: vequincozes@midiacom.uff.br (V.E. Quincozes), silvioquincozes@unipampa.edu.br (S.E. Quincozes), celio@ic.uff.br (C. Albuquerque), diego.passos@isel.pt (D. Passos), mosse@pitt.edu (D. Mossé).

https://doi.org/10.1016/j.cose.2026.105217
Received 11 April 2025; Received in revised form 16 October 2026; Accepted 17 October 2026
Available online 24 October 2026
0167-4048/© 2026 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).
The Article Processing Charge for the publication of this research was funded by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior - Brasil (CAPES) (ROR identifier: 00x0ma614).

Acknowledgments 16
Data availability 16
References 16

1. Introduction

Integration of computing and communication into physical equipment processes brought to light a new paradigm called Cyber–Physical Systems (CPSs). These systems are being employed in several critical infrastructure applications, including water treatment (Goh et al., 2016) and distribution systems (Taormina et al., 2018), power generation, transmission, and distribution (Quincozes et al., 2021), and monitoring systems (Hink et al., 2014).

In contrast to traditional Information Technology (IT) systems, in which confidentiality and integrity typically have higher priority than availability, CPSs have hard requirements of timely results and real-time communication. Also, because CPS applications have historically been disconnected from the Internet, they were originally only vulnerable to attacks originating from within the facilities. Thus, these applications often use standards and protocols that lack proper security defenses against outside agents (Chui et al., 2023). The networked nature of CPS opens up the possibility of many new interconnected applications but also creates widespread potential vulnerabilities across critical infrastructure.

These vulnerabilities include weaknesses in remote access mechanisms and local attacks, such as the historical Stuxnet malware, which targeted Iran's nuclear facilities in 2010 by compromising the control systems of uranium enrichment centrifuges (Falliere et al., 2011). Therefore, it is essential to design advanced security defenses to deal with a range of cyberattacks that target CPSs (Quincozes et al., 2021).

In this context, Intrusion Detection Systems (IDSs) play a vital role in detecting malicious activities across diverse CPS applications. Although widely used, designing and deploying IDSs is complex, especially in very specialized domains where specific communication protocols and devices demand tailored solutions (Premaratne et al., 2010). To design an IDS, it is necessary to decide which data source will be analyzed, the detection coverage, the evaluation strategy, and the performance metrics (Quincozes et al., 2021). Given the diversity of CPS applications, we will use "CPS security" as a general term to encompass all domains, though each — such as smart grid or water distribution security — requires domain-specific IDS adaptations. These adaptations may reflect not only protocol differences, but also distinct process dynamics and operational constraints across CPS domains. For example, smart grids are commonly associated with protection and automation functions under stringent timing requirements (Wollman et al., 2017; Quincozes et al., 2021), whereas water and pipeline systems are more directly tied to process-state variables such as flow, pressure, level, and actuator status (Transportation Security Administration, 2018; Gulzar and Mustafa, 2024). Unmanned Aerial Vehicles (UAVs), in turn, place stronger emphasis on wireless links, telemetry, and navigation-related data (Mykytyn et al., 2024).

In particular, IDSs developed for CPSs nowadays are based on Machine Learning (ML) techniques that either learn the baseline of normal behavior to flag deviations (*e.g.*, zero-day attacks) or rely on labeled datasets to identify features of known attacks. However, gathering data for either approach poses challenges: due to the criticality and sensitivity of CPSs, data about attacks in these scenarios is typically not publicly shared, limiting the development of IDSs for CPSs. Ideally, datasets would be extracted directly from real systems to provide a comprehensive representation of threats. In practice, however, due to the scarcity of such datasets, synthetic dataset generators serve as a convenient starting point, though they fail to fully represent the wide variety of CPSs and their corresponding attack scenarios. As a result, current efforts to devise effective IDSs for CPSs are constrained by the limited availability of relevant training data (Quincozes et al., 2021; Tushkanova et al., 2023). Furthermore, even among the available domain-specific datasets, there remains a gap in the literature: the lack of a structured basis to categorize, compare, and assess how well datasets represent cyber and physical features as well as attack coverage across CPS domains. To address this gap, this paper focuses on the analysis of labeled datasets for classifier-based IDSs and on providing insights that may be valuable for anomaly-based IDSs.

Our main contributions include (i) a new taxonomy of CPS datasets, categorizing them based on key attributes such as data sources and types of attacks represented therein; (ii) a thorough literature review on the available intrusion detection datasets for CPSs; (iii) an analysis of the data in existing datasets; (iv) a detailed discussion on the challenges — such as the need for high-quality data and dataset suitability — and opportunities, like developing domain-specific datasets and applying feature enrichment and selection techniques; and (v) an experimental validation of the proposed taxonomy through the introduction of a novel open-source Standardized Evaluation Framework (SEF-CPS), which systematically classifies datasets, exposes key limitations (e.g., imbalanced attack distributions and limited feature diversity), and enables more structured IDS experiment design.

The remainder of this work is organized as follows. In Sections 2 and 3, we propose a new taxonomy, discuss related work, and the challenges of CPS intrusion detection, respectively. In Section 4, we review in detail the available datasets for building IDSs in CPSs and present insights on how to improve the feature representativeness in the currently available datasets. In Section 5, we present experiments that validate the taxonomy by applying it to publicly available datasets via the reproducible SEF-CPS framework, demonstrating how it systematically categorizes them, uncovers representational gaps, and informs improvements in dataset quality for intrusion detection. Finally, Section 6 presents our final remarks and conclusions.

2. Taxonomy and related work

We present a taxonomy and other aspects of datasets for CPSs in Fig. 1. We devised this taxonomy to categorize existing datasets based on attributes that are essential for CPS security research, such as data sources, feature types, applications, and types of attacks. In Section 4, we summarize the datasets according to the proposed taxonomy.

To contextualize our taxonomy within the existing literature, we conducted a literature search in Google Scholar covering the years 2019 to 2026. Our search strategy was iterative and based on core concepts: the broad application domain ("cyber–physical systems" or "CPS"), the research object (e.g., "dataset", "data sets"), and the security context (e.g., "security", "cybersecurity", "intrusion detection"). Specific CPS sub-domains (such as smart grids, water treatment, and gas pipelines) were not restricted *a priori*; rather, they were naturally identified and categorized as a result of the literature retrieval. Furthermore, we expanded our initial selection using forward and backward citation tracking (snowballing).

This search served a dual purpose: mapping related work and identifying the primary datasets for our study. For the related work analysis (summarized in Table 1), we filtered the papers based on specific Inclusion (IC) and Exclusion (EC) Criteria. We included (IC1) studies published between 2019 and 2026, (IC2) works written in English, (IC3) surveys, reviews, and targeted studies that address general CPS security or specifically discuss cybersecurity datasets, and (IC4) peer-reviewed articles published in recognized journals, conferences, or book chapters. As an exception to IC4, we included (Tantawy, 2022) due to its specific focus on the architectural elements of CPS datasets.

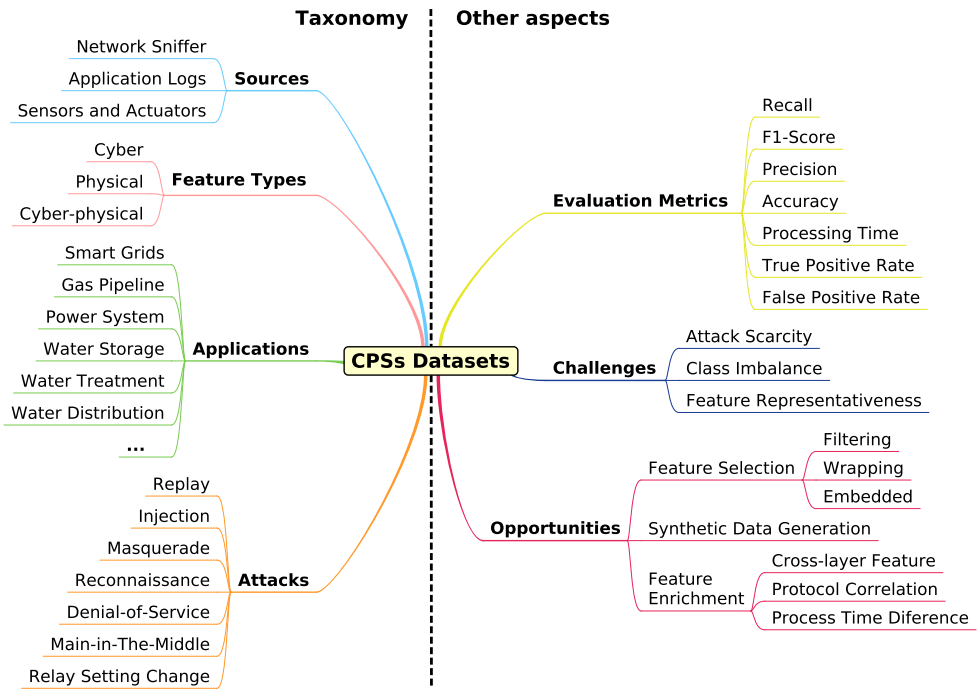


Fig. 1. CPS dataset proposed taxonomy.

Table 1
Related work.

Ref.	Year	Dataset taxonomy	Sources	Feature types	Applications	Attacks	Evaluation metrics	Challenges	Opportunities
Xu and Duan (2019)	2019	✗	✗	✗	β	β	✗	✓	✓
Lin et al. (2020)	2020	✗	✗	✗	✗	✗	✗	✗	✗
Khalid et al. (2020)	2020	✗	✗	✗	✓	✓	✗	✗	✗
Zhang et al. (2021)	2021	✗	✗	✗	✓	✓	✓	✓	✓
Conti et al. (2021)	2021	✗	✓	✓	✓	✓	✓	✓	✓
Wang et al. (2021)	2021	✗	✗	✗	✓	✓	✗	✗	✗
Olowononi et al. (2021)	2021	✗	✗	✗	✓	✓	✓	✓	✓
Faheem and Butt (2022)	2022	✗	✗	✗	β	✗	✗	✗	✗
Snehi and Bhandari (2022)	2022	✗	✗	β	β	✓	β	✓	✓
Alshaibi et al. (2022)	2022	✗	✗	✗	β	✓	✗	✓	✓
Tantawy (2022)	2022	✗	✓	✓	✓	✓	✗	✓	✓
Kayan et al. (2022)	2022	✗	✗	β	✓	✓	✓	✓	✓
Alwan et al. (2022)	2022	✗	✗	✗	✓	✗	✗	β	β
Tushkanova et al. (2023)	2023	✗	β	✗	✗	β	✓	✗	✗
Jamal et al. (2023)	2023	✗	✗	✗	β	β	β	✓	✓
Hutzelmann et al. (2023)	2023	✗	✗	✗	✗	✗	✗	✗	✗
Sheikh and Singh (2023)	2023	β	✗	β	✗	✓	✓	✓	✗
V.E. Quincozes et al. (2024)	2024	✗	✗	✗	✓	✓	✓	✓	✓
Gulzar and Mustafa (2024)	2024	✗	β	β	β	✓	✗	β	β
Ji et al. (2024)	2024	✗	β	β	✓	✓	✓	✓	✗
Goldschmidt and Chudá (2025)	2025	✓	β	β	✗	✓	✗	✓	β
Ashraf and Masoodi (2026)	2026	✓	β	β	✓	✓	✓	✓	✓
Vigneshwaran et al. (2026)	2026	β	β	β	β	✓	✓	✓	✓
Reddy and Kumar (2026)	2026	✓	β	β	β	✓	✓	✓	✓
Zhou et al. (2026)	2026	β	✓	✓	β	✓	✓	✓	β
Termanini et al. (2026)	2026	✓	✓	✓	✓	✓	✓	✓	✓
This	2026	✓	✓	✓	✓	✓	✓	✓	✓

Legend for cells: (β) Briefly mentioned; (✓) Covered in the manuscript; (✗) Not covered.

We excluded (EC1) grey literature, such as monographs, theses, and dissertations, and (EC2) primary studies that solely proposed new intrusion detection models without providing a broader literature review or an in-depth characterization of datasets.

Table 1 summarizes related work, providing an overview of literature on datasets for CPS security. Each row represents a study, and each column addresses a specific aspect related to CPS datasets. To ensure objectivity, we defined criteria for each classification: a check mark (✓) indicates the presence of a dedicated section or an extensive discussion on the topic; a beta symbol (β) denotes that the aspect is briefly mentioned or discussed in only a few paragraphs without a dedicated structure; and a cross (✗) indicates the absence of the topic. Specifically,

the Challenges column identifies whether the study provides a structured analysis of the limitations, hurdles, or open problems currently faced by CPS datasets (e.g., lack of physical-layer representativeness). The Opportunities column reflects whether the work proposes or discusses concrete future directions or innovative approaches to improve dataset quality, structure, or detection effectiveness (e.g., cross-layer feature enrichment). There are numerous reviews, overviews, surveys, and studies on the security of cyber–physical systems, yet they often do not specifically focus on datasets (Olowononi et al., 2021; Wang et al., 2021; Jamal et al., 2023; Khalid et al., 2020; Zhang et al., 2021). These studies typically limit themselves to general security discussions without delving into the specifics of the data used. Other works address challenges of data quality and privacy at large scale for CPS (Alwan et al., 2022; Xu and Duan, 2019; Faheem and Butt, 2022). While those provide valuable insights into large-scale data management, they do not detail the specifics of CPS datasets.

Some studies discuss datasets in specific contexts. For example, Hutzelmann et al. (2023) focus on generating tailored and confined datasets for the evaluation of IDSs in CPSs, highlighting the need for specialized datasets in safety-critical environments. Similarly, Lin et al. (2020) examine datasets from the iTrust group, which are used for cybersecurity research and educational purposes, particularly in experiments involving Industrial Control Systems (ICS). Gulzar and Mustafa (2024) provide an analytical survey on CPS security in water treatment and distribution, evaluating intrusion detection techniques and dataset limitations in this domain. While these contributions are vital, they focus on specific applications, which may limit the generalizability of their findings. In contrast, our work provides a broader perspective, analyzing datasets and IDS evaluation across multiple CPS domains rather than being confined to a single application context.

Some works mention datasets in CPSs (Alshaibi et al., 2022; Ji et al., 2024; Snehi and Bhandari, 2022; Tushkanova et al., 2023). However, Alshaibi et al. (2022) do not explicitly mention evaluation metrics, Snehi and Bhandari (2022) focus more on Internet of Things (IoT) datasets, and Tushkanova et al. (2023) do not discuss data source issues, feature types, or challenges and opportunities. Also, Ji et al. (2024) propose taxonomies for IDSs and attacks in CPSs, offering a broad discussion on IDS techniques, industrial protocols, and security challenges. Although they mention datasets such as KDDCup99 and Bot-IoT, their work does not focus on categorizing or structuring datasets specifically but rather focuses on attacks and detection. Similarly, Sheikh and Singh (2023) provide a broad survey on IDSs and machine learning models applied to CPS security, analyzing existing datasets and discussing common evaluation metrics. However, like Ji et al. they do not propose a structured taxonomy for datasets nor validate dataset-related challenges through experimentation. Additionally, most of the datasets they review are either outdated or focused on traditional Information Technology (IT) and IoT networks, whereas our work incorporates recent CPS-specific datasets such as ERENO and PowerDuck.

More recently, Goldschmidt and Chudá (2025) proposed a taxonomy for network intrusion datasets, analyzing 89 public datasets based on 13 properties. While their study provides a broad classification, it is not specifically designed for CPS datasets and does not consider physical components or sources unique to cyber–physical environments. Additionally, although they review widely used datasets such as NSL-KDD and DARPA 1999, they do not include more recent CPS-specific datasets like ERENO, PowerDuck, SwaT.

Finally, some studies cover various aspects in CPS security datasets, analyzing their sources, attack types, and limitations. While Conti et al. (2021) presents a review of ICS testbeds and datasets, identifying their key characteristics and security challenges, Tantawy (2022) discusses the essential elements that should be present in CPS security datasets and suggests a dataset architecture. However, neither study evaluates datasets through experimental analysis nor explores techniques for dataset enhancement. To address these gaps, our work not only investigates more recent datasets (2021–2025) but also examines how dataset characteristics impact IDS performance through experimental validation.

Recent studies have examined intrusion detection datasets from different perspectives. Ashraf and Masoodi (2026) surveyed 124 datasets for communication networks, including IoT, Industrial IoT (IIoT), and CPS contexts, while Reddy and Kumar (2026) investigated dataset realism and evaluation biases in IoT/IIoT intrusion detection. Vigneshwaran et al. (2026) reviewed attack detection methods and benchmark datasets specifically for smart grids, whereas Zhou et al. (2026) introduced ICS-NAD, an ICS network traffic dataset covering 20 attack types. More closely related to our study, Termanini et al. (2026) characterized 23 ICS/IIoT datasets through a multidimensional taxonomy distinguishing network traffic from physical process data and proposed criteria for documentation completeness and dataset suitability. However, these studies do not combine CPS-specific dataset characterization with experimental comparisons of the separate and combined contributions of cyber and physical features to intrusion detection across multiple datasets.

Going beyond these works, our study introduces a structured taxonomy specifically for CPS intrusion datasets, addressing gaps in how these datasets are categorized based on key attributes such as data sources, attack coverage, and feature representation. We validate the practical utility of our taxonomy using the proposed open-source SEFCPS framework to evaluate intrusion detection models across seven CPS datasets. We further compare cyber-only, physical-only, and combined feature sets across three datasets and apply explainability analysis to examine the contribution of individual cyber and physical features to model predictions.

3. CPS intrusion detection dataset challenges

The fundamental differences between CPS and traditional IT systems in intrusion detection arise mainly due to the particular protocols and physical devices involved. CPSs integrate both physical and cyber components, resulting in different attack vectors, *modus operandi*, and attacker behavior (Rocchetto and Tippenhauer, 2016). Therefore, the abundant data available for general-purpose IDS is not suitable for CPS-based IDS. Nowadays, many IDSs use ML algorithms to build an IDS model, and it is necessary to train them with the appropriate attacks — and often benign traffic.

3.1. The need for data

ML approaches to identify suspicious behavior related to cyberattacks represent the network behavior through a set of features. When analyzed together, these features can reveal potential threats. The major challenges faced by ML-based IDSs designed for CPSs include acquiring realistic traffic information about the application domain and up-to-date attack models that showcase the attacker's behavior.

There are two main approaches for implementing an IDS to detect intrusions: *behavior-based* or *knowledge-based*. While the former approach focuses on profiling the normal user actions in the CPS, the latter focuses on mapping the well-known attacks into a signature database (Quincozes et al., 2021). Both approaches critically depend on the availability of comprehensive and representative data.

The behavior-based approach does not attempt to understand or discover specific attack classes or attack signatures. Instead, it focuses on detecting changes from baseline behavior (*i.e.*, a departure from the *expected* or *normal* profile). One disadvantage of this approach is its susceptibility to false positives, which unnecessarily trigger defenses (Quincozes et al., 2021). Another disadvantage is that, if there were attacks during the training phase, they would be considered as part of the expected normal CPS/network behavior. As these systems typically use unsupervised data to establish such thresholds, it is hard to distinguish between a biased model and a proper model. Also, the

benign network behavior may change over time and these legitimate changes may trigger false alarms.

Signature-based (or knowledge-based) IDSs rely on a previously existing dataset that contains signatures representing both normal and attack profiles. This method, also known as *misuse detection* (Ahmad et al., 2021), is highly effective in reducing false positives and providing accurate results by matching system activities against known attack signatures rather than modeling expected behavior in production CPSs. However, it is strongly dependent on the availability of comprehensive and up-to-date intrusion detection datasets.

Therefore, performing intrusion detection by using a signature-based IDS can support more precise detection. This approach is preferable whenever an intrusion detection dataset is available for the CPSs domain.

3.2. Dataset suitability

One of the key differences between IT systems and CPSs is the nature of the collected information. Whereas traditional IT systems are limited to host- or network-based information, CPS applications monitor physical processes. Physical processes introduce an additional layer of information, called the physical process layer, which is potentially a target of benign or malicious actions. Furthermore, specific CPSs have proprietary communication protocols that can be exploited in unique ways (*e.g.*, they could cause protective relays to misinterpret system states, potentially leading to a blackout) (S.E. Quincozes et al., 2024; Lin et al., 2021). Thus, new datasets with attack vectors and targeted towards domain-specific CPSs (not general IT systems) are needed.

Additionally, CPSs face the challenge of supporting legacy protocols, where heterogeneous communication protocols may co-exist due to the high cost of replacing legacy devices. For example, digital substations have integrated Ethernet-based protocols like the Generic Object Oriented Substation Event (GOOSE) and Sampled Values (SV) protocols, replacing several serial-based communication protocols (Quincozes et al., 2021). Another example is Modbus in water distribution systems (Katulić et al., 2023) and DNP3 in power grids (Qassim et al., 2023); both were designed for isolated environments, prioritizing simple and rapid communication over security. However, their lack of native encryption and authentication made them vulnerable to interception, command injection, and other attacks. This complexity and diversity in communication protocols demand that IDSs cover all existing systems effectively. Thus, datasets must encompass a wide range of protocols and ensure comprehensive coverage.

Finally, because they are embedded in physical processes, applications of CPSs have a direct impact on the physical world. Consequently, ensuring the security of CPSs is crucial, as a successful attack can have severe real-world consequences, including putting human lives at risk. This underscores the necessity for datasets that accurately reflect the unique security challenges and potential attack vectors specific to CPS environments. For instance, the HAI dataset (Shin et al., 2020), used in critical infrastructure like water treatment systems, simulates physical attack scenarios that could lead to catastrophic failures in water supply. Similarly, the ERENO dataset (S.E. Quincozes et al., 2024) is applied for detecting cyber-attacks in smart grids, which are essential for managing and distributing electricity. The ERENO dataset includes attack scenarios such as Masquerade Attacks, where attackers report normal operations during a fault to cause equipment damage, potentially causing significant disruption to grid operations.

3.3. Data generation and collection

CPSs comprise specialized devices, often expensive, designed for particular functions (*e.g.*, measurement and actuation in a power grid, advanced medical devices, gas pipelines and valves, and water distribution systems). Replicating this kind of environment in a lab to generate appropriate datasets is generally infeasible. Moreover, CPSs often carry confidential and sensitive data.

The scarcity of domain-specific CPS datasets causes many authors of CPS IDSs to still use traditional (IT) datasets for their training and evaluation (Tantawy, 2022; Tushkanova et al., 2023). One alternative to address this concern is to adopt synthetic data generators tailored to CPS domains. These tools simulate CPS environments and generate data that mimic real-world scenarios, allowing researchers to create datasets for IDS training and evaluation. However, synthetic data generation has its limitations. These tools typically focus on a limited set of operations or very specific functions of a CPS, which may not fully capture the diverse range of cyberattack scenarios and could limit the effectiveness of IDSs (Tushkanova et al., 2023). Additionally, while synthetic data can be useful for preliminary testing and development, it may not always reflect the complexity and unpredictability of real-world CPS environments.

Therefore, current state-of-the-art data for IDS training and assessment comprise public datasets (*e.g.*, Faramond Faramondi et al., 2021, and PowerDuck Zemanek et al., 2022), simulation tools (*e.g.*, COSSIM Zahariadis and Papaestathiou, 2022), and traffic generation frameworks (*e.g.*, Framework S.E. Quincozes et al., 2024), as detailed in Section 4.

4. Intrusion detection datasets

According to the taxonomy presented in Fig. 1, CPS datasets can be categorized based on several key attributes, including data sources, feature types, applications, and types of attacks. In this section, we summarize the currently used datasets according to the proposed taxonomy, discussing their characteristics and categorization. A synthesized overview of their key characteristics is presented at the end of the section, facilitating comparison between datasets.

4.1. Applications and attacks

We found datasets for various CPS applications, including safe water treatment (Goh et al., 2016), water distribution (Taormina et al., 2018; Faramondi et al., 2021; Ahmed et al., 2017), and water storage (Morris and Gao, 2014), liquid storage and distribution (Laso et al., 2017), unmanned aerial vehicles (Hassler et al., 2023), gas pipelines (Morris and Gao, 2014; Morris et al., 2015; Filonov et al., 2016), and smart grids (S.E. Quincozes et al., 2024; Sahu et al., 2021; Adepu et al., 2019; Biswas et al., 2019; Zemanek et al., 2022). The objective of each application can be summarized as follows.

- **Water Treatment.** Treats the water before making it available for consumption. To generate a suitable dataset, Goh et al. (2016) used a scaled-down replica of city water treatment plants capable of producing 5 gallons/minute of double filtered water to simulate the entire treatment process: from raw water capture to storage and distribution.
- **Water Distribution.** Ensures water reaches the population effectively. Consists of a network formed by a set of pipes, junctions, storage tanks, pumps, valves, and reservoirs that distribute water to the population (Taormina et al., 2018).
- **Water storage.** Manages water levels for availability and quality. Composed of tanks, valves, sensors, and pumps that allow monitoring and adjusting of the water levels, supporting the distribution network (Morris and Gao, 2014).
- **Liquid Storage and Distribution.** Provides infrastructure for the safe storage and distribution of various liquids. Datasets include different systems for storing and distributing liquids, covering both control and operational security aspects (Laso et al., 2017).
- **Gas pipeline.** Transports gas safely through controlled systems. Consists of hermetic piping networks, compressors, pressure gauges, and valves that allow gas transport (Morris and Gao, 2014).

Table 2
Categorization of the current CPSs intrusion datasets, where S&A: Sensors and Actuators, NS: Network Sniffer, AL: Application Logs, C: Cyber Features, P: Physical Features, DoS: Denial of Service, MitM: Man-in-the-Middle.

Application	Dataset	Source	Attacks	Feature
Water treatment	SWaT (Goh et al., 2016)	S&A, NS	Injection	C, P
Water distribution	WADI (Ahmed et al., 2017)	S&A	Spoofing	C, P
	BATADAL (Taormina et al., 2018)	S&A	Replay	C, P
	Faramond (Faramondi et al., 2021)	S&A, NS	DoS, MitM, Scanning, Physical	C, P
Water storage	Morris-3 (Morris and Gao, 2014)	NS, AL	Injection, Reconnaissance, DoS	C, P
Water/Fuel storage & Distribution	Laso (Laso et al., 2017)	Sensors, NS	DoS, Spoofing, Physical	C, P
Gas pipeline	Morris-2 (Morris and Gao, 2014)	NS, AL	Injection, Reconnaissance, DoS	C, P
	Morris-4 (Morris et al., 2015)	NS, AL	Injection, Reconnaissance, DoS	C, P
	GHL (Filonov et al., 2016)	S&A	Control/Command Manipulation	C, P
Smart grids	Morris-1 (Hink et al., 2014)	NS, AL	Injection, Relay Setting Change	C, P
	ERENO (S.E. Quincozes et al., 2024)	NS	Injection, Masquerade, Replay, DoS	C, P
	EPIC (Adepu et al., 2019)	S&A	MitM, Malware Attack and Physical Damage	C, P
	Biswas (Biswas et al., 2019)	NS, AL	DoS, Message Suppression, Data and Control Manipulation	C, P
	PowerDuck (Zemanek et al., 2022)	NS, AL	Replay, Insertion, Suppression, Flooding	C, P
	Sahu (Sahu et al., 2021)	NS, AL	Injection, MitM, False Data	C, P
Unmanned aerial vehicles	Hassler (Hassler et al., 2023)	S&A, NS, AL	De-Authentication (DoS), Replay, Injection, Evil Twin	C, P
Industrial control system	ICS-Flow (Dehlaghi-Ghadim et al., 2023)	NS, AL	Reconnaissance, Replay, DDoS, MitM	C
	Westermo (Strandberg et al., 2023)	NS, AL	Harmless/Bad SSH, Misconfigured/Duplicated IP address, Port-Scan, and MitM	C, P
	Electra (Gómez et al., 2019)	NS, AL	Reconnaissance, Injection, Replay	C
	HAI 1.0 (Shin et al., 2020)	S&A	Control Manipulation and Stealthy Attacks	C, P
	ICS-ADD (Gaggero et al., 2024)	NS, AL	Reconnaissance, MitM, Injection, Brute-force, DNS Tunneling, DoS	C, P

- **Smart Grids.** Provides more efficient and secure power grids than traditional ones. They are formed by circuit breakers, relays, power generators, and Intelligent Electronic Devices (IEDs), among other devices. IEDs, in particular, execute control and protection tasks, analyzing electrical parameters and taking actions, such as turning circuit breakers off in case of faults (Hink et al., 2014).
- **Unmanned Aerial Vehicles (UAVs).** Ensures secure operation of autonomous aerial systems. UAV security datasets capture network and sensor data from drones to detect cyber–physical attacks (Hassler et al., 2023). While not traditionally classified as critical infrastructure, UAVs are increasingly used in security, inspection, and emergency response, where cyber–physical threats can disrupt operations.
- **Industrial Control Systems (ICS).** Supports operational continuity and security across CPS environments. Although not strictly CPS applications, ICS components, such as Supervisory Control and Data Acquisition (SCADA) systems and Programmable Logic Controllers (PLCs), are critical in managing and automating processes in utilities and infrastructure. Datasets for ICS focus on security and anomaly detection to ensure reliability in critical infrastructure (Dehlaghi-Ghadim et al., 2023; Strandberg et al., 2023; Gómez et al., 2019; Shin et al., 2020; Gaggero et al., 2024).

These applications are large systems (*e.g.*, to distribute water or electricity to an entire city) with multiple critical components, and therefore, when they become connected to the Internet, the risk of attacks becomes extremely important. There are two main types of attacks: cyber and physical. Cyberattacks are executed over a network to disrupt or manipulate the behavior of a system, infrastructure, or device remotely. In CPSs, these attacks may target the actual hardware components (*e.g.*, a valve or an IED) with the intent to damage, disable, or make them operate incorrectly (Goh et al., 2016). In this paper, we only consider cyberattacks and not physical attacks to the CPSs.

Table 2 presents a categorization of the current CPS intrusion datasets, summarizing their applications, data sources (representing how data is collected, such as from sensors, network sniffer, or application logs), types of attacks, and feature types. There are several types of cyberattacks, such as Data Injection (Goh et al., 2016), Replay (Taormina et al., 2018), Masquerade (S.E. Quincozes et al., 2024), Denial-of-Service (DoS) (S.E. Quincozes et al., 2024), Relay Setting Change (Morris and Gao, 2014), Reconnaissance (Morris et al., 2015), and Man-in-the-Middle (MitM) (Faramondi et al., 2021). In particular, data injection attacks are the most common and can affect most applications, including water treatment (Goh et al., 2016), water storage (Morris and Gao, 2014), gas pipeline (Morris and Gao, 2014; Morris et al., 2015) and smart grid (S.E. Quincozes et al., 2024; Hink et al., 2014; Sahu et al., 2021). In this type of attack, an attacker inserts malicious messages into the network to deceive the control system, disrupting its observability of a process. For example, in a water treatment plant, a false data injection could involve sending fake sensor readings that indicate a low water level to the controller, when the actual level is within safe limits. This can cause the system to react unnecessarily by activating pumps or shutting down valves, potentially destabilizing the entire process.

It is important to note, however, that the temporal dynamics, observability, and physical impacts of these attack classes vary significantly across CPS domains. In smart grids, for instance, attacks are often discussed together with timing-sensitive protection and automation workflows (Wollman et al., 2017; Quincozes et al., 2021). Injection attacks targeting IEC-61850 protocols (e.g., GOOSE and SV) operate within strict latency requirements; a manipulated message can trip a circuit breaker in a matter of milliseconds, potentially triggering instantaneous cascading failures (S.E. Quincozes et al., 2024). In contrast, the thermodynamic and hydraulic processes in water infrastructures exhibit higher physical inertia and are more often interpreted through process evolution and control-state inconsistencies (Gulzar and Mustafa, 2024). A false sensor reading in these environments may take minutes or hours

to manifest as a critical physical failure (e.g., a tank overflow) (Goh et al., 2016; Taormina et al., 2018). Furthermore, in UAVs, communication, telemetry, and navigation links form a more prominent part of the attack surface (Mykytyn et al., 2024), while in pipelines, cybersecurity guidance places particular emphasis on control-system integrity and operational resilience (Transportation Security Administration, 2018).

Beyond these differences in manifestation and impact, the scope of injection attacks on CPSs is potentially large. A point of attack can be a physical element or a communication network that connects sensors and actuators to controllers. For example, the SWaT dataset (Goh et al., 2016) has six stages, each with different sensors and actuators. These attacks can be concentrated on (i) just one point of a CPS, (ii) on two or more points but only on one stage, (iii) on more than one element of any one stage, or (iv) on two or more stages of the CPS (Goh et al., 2016).

Two other attacks commonly included in datasets for water and gas pipeline storage applications are Reconnaissance and DoS. Reconnaissance attacks aim to gather information from the SCADA system — an industrial control system used to monitor and control the infrastructure — and to map the network to identify device characteristics, while DoS attacks deplete communication or system resources (Morris and Gao, 2014). Although DoS attacks may be easier to detect, they are challenging to prevent in CPSs due to limited device resources, reliance on legacy protocols, and a lack of redundant communication paths, making these systems especially vulnerable to disruptions. For example, in smart grids, DoS attacks can disrupt the communication between control centers and substations, affecting the stability of power distribution. In relay setting change attacks, used in smart grids, attackers intercept communication between two agents and, instead of viewing or modifying the communication, they simply relay it to another device. Man-in-the-Middle attacks, present in datasets for water distribution (Faramondi et al., 2021) and power systems applications (Sahu et al., 2021), are used by malicious users to capture and modify messages during the communication between two legitimate parties.

Unlike other CPSs, UAVs rely entirely on wireless communication, making them particularly susceptible to network-layer attacks. In addition to traditional cyber threats, UAVs are uniquely vulnerable to attacks such as Evil Twin (a form of MitM) and De-Authentication (a type of DoS) (Hassler et al., 2023). Evil Twin attacks occur when an attacker sets up a rogue Wi-Fi access point that mimics a legitimate one, tricking the UAV into connecting to it and allowing interception or manipulation of data. De-Authentication attacks, on the other hand, exploit the 802.11 protocol to forcibly disconnect the UAV from its network, disrupting its communication and making it susceptible to takeover or loss of control, potentially leading to mission failure or drone loss.

Table 3 shows the number of types of attacks, samples, features, the percentage of the dataset (from Table 2) that are attack samples (with respect to the total number of samples), and the data source. It is interesting to note that those datasets vary significantly in both overall size and the number of different types of attacks. In the data source column, we categorize datasets into two groups: (i) Testbed, created using physical equipment in lab settings (*e.g.*, SWaT, which includes pumps, sensors, and PLCs), and (ii) Simulated, using virtual tools without physical devices, like the Power Systems Computer-Aided Design (PSCAD) tool (Ltd, 2022) to simulate realistic electrical signals for smart grids. None of the datasets we found were collected in large-scale field deployments.

The Sahu dataset (Sahu et al., 2021) is not publicly available, hindering our ability to obtain specific details. Furthermore, we encountered challenges in precisely determining the attack percentages for the Westermo (Strandberg et al., 2023), GHL (Filonov et al., 2016), and HAI (Shin et al., 2020) datasets, due to the lack of explicit documentation and data in the publicly available files.

It is also important to highlight the technical specificity of certain datasets that contribute to the broader understanding of CPS security challenges. For example, the ICS-Flow (Dehlaghi-Ghadim et al., 2023) and Westermo (Strandberg et al., 2023) datasets, each with 64 features (two labeled in a binary format – normal or malicious – and two in a multiclass format providing attack name), were generated for ICS applications using the same simulator (*i.e.*, ICS-FlowGenerator). These datasets employ dual labeling strategies to address different aspects of anomaly detection: *Injection Timing* labels all traffic during attack periods as malicious, while *Network Security Tools* focuses only on traffic directly involving any attacker. What differentiates those last two datasets is that they encompass distinct attacks and number of samples, underscoring the variability and targeted nature of CPS attack scenarios.

In terms of features, the Electra dataset (Gómez et al., 2019) is comprised of 11 features, including one label, showcasing a more streamlined feature set. Similarly, HAI 1.0 (Shin et al., 2020) enriches the dataset landscape with 64 features, of which 4 serve as labels that indicate the presence of an attack, with the first label denoting a system-wide attack and the subsequent labels indicate attacks on specific processes, offering a view of potential security vulnerabilities in ICS to CPS environments. Adding to the diversity of dataset structures, the Laso (Laso et al., 2017) dataset presents an even more constrained approach, featuring only 3 features (*i.e.*, timestamp, register, and value).

Other datasets, such as Biswas et al. (2019) and Gaggero et al. (2024), provide network traffic data in PCAP (Packet Capture) format, which stores raw packet-level information transmitted over a network. PCAP files capture the full structure of network communications, including headers, payloads, timestamps, and protocol details, allowing for deep traffic analysis. However, they do not provide a predefined set of extracted features or directly labeled attack instances, making them challenging for direct use in artificial intelligence applications. Their utility depends on preprocessing steps, such as packet parsing, feature extraction (e.g., flow statistics, timing characteristics, and protocol behaviors), and manual or automated labeling to identify attack scenarios (Hussain et al., 2021). This necessity adds a layer of complexity, as researchers must convert the raw packet data into a structured format suitable for ML models.

The PowerDuck (Zemanek et al., 2022) dataset exhibits flexibility in its structure by incorporating a ‘data’ field whose length varies across samples. While it consistently presents 11 common fields, including identifiers such as ‘id’, ‘timestamp’, ‘protocol’, and ‘malicious’ indicators, the ‘data’ field incorporates different measurements and commands, reflecting the diversity of data points collected in the SCADA systems. This variability is influenced by the specific types of IEDs being monitored and the operational context, which may include various parameters such as voltage, current, or control commands. Moreover, the inclusion of sequence-related features such as ‘StNum’ and ‘SqNum’ suggests an emphasis on temporal and event sequencing, which is critical for understanding the behavior of CPSs under various scenarios, including attack detection.

Another dataset of interest is UAV Hassler (Hassler et al., 2023), which includes both cyber and physical features, making it relevant for CPS intrusion detection. Unlike most datasets that integrate different data types within a single entry, Hassler stores cyber and physical data in separate sections with distinct feature sets. Cyber samples contain 37 features, while physical samples have 16 features, and there is no direct alignment between them. Ideally, datasets of this nature should provide a unified structure where cyber and physical features are combined at the instance level, allowing for a more comprehensive analysis of cross-domain attack patterns. However, it is important to acknowledge that in practical scenarios, these data sources are often collected asynchronously and at different sampling rates. Therefore, the integration of cyber and physical data at the IDS level poses challenges, as the method of aggregation itself can influence detection performance (Al-Ghuwairi et al., 2023). Addressing this challenge is crucial to ensuring

Table 3
Characteristics of existing CPS datasets.

Dataset	Attack types	Number of samples	Features	Attack percentage	Data source
SWaT (Goh et al., 2016)	1	946,722	51	5.87%	Testbed
WADI (Ahmed et al., 2017)	15	172,803[a]	130	5.77%	Testbed
BATADAL (Taormina et al., 2018)	14	12,940	45	1.69%	Simulated
Faramond (Faramondi et al., 2021)	4	24,306,718	56	68.08%	Testbed
Morris-3 (Morris and Gao, 2014)	7	236,179	23	27.00%	Testbed
Laso (Laso et al., 2017)	3	337,438	3	30.98%	Testbed
Hassler (Hassler et al., 2023)	4	54,784	53	74.97%	Testbed
Morris-2 (Morris and Gao, 2014)	7	97,019	26	36.96%	Testbed
Morris-4 (Morris et al., 2015)	35	274,628	16	21.87%	Testbed
GHL (Filonov et al., 2016)	4	11,356,701	23	–[b]	Simulated
Morris-1 (Hink et al., 2014)	40	76,035	128	77.85%	Testbed
ERENO (S.E. Quincozes et al., 2024)	7	5,911,296	69	7.28%	Simulated
EPIC (Ahmed and Kandasamy, 2021; Adepu et al., 2019)	3	5230	292 - 391[c]	–[b]	Testbed
Biswas et al. (2019)	3	344,858	PCAP	89.77%	Simulated
PowerDuck (Zemanek et al., 2022)	4	1,133,589	Varies	97.62%	Testbed
Sahu (Sahu et al., 2021)	3	–[d]	28	–[d]	Simulated
ICS-Flow (Dehlaghi-Ghadim et al., 2023)	4	45,719	64	33.86% & 19.71%[e]	Simulated
Westermo (Strandberg et al., 2023)	6	68,730	64	21.39% & 65.30%[e]	Testbed
Electra (Gómez et al., 2019)	3	16,289,279	11	4.42%	Testbed
HAI 1.0 (Shin et al., 2020)	3	995,404	64	1.83%	Testbed
ICS-ADD (Gaggero et al., 2024)	7	215,456	PCAP[f]	–[f]	Testbed

[a] WADI has two datasets: one with 784,572 samples, unlabeled for attacks, and another with 172,803 labeled samples. We prioritize the labeled dataset for our analysis.
[b] Sample labeling was not found, preventing identification of attack percentages.
[c] Scenarios 1–6 have 292 features, while scenarios 7 and 8 have 391.
[d] Information on the number of samples and attack percentages could not be computed directly.
[e] Two attack percentages are due to distinct labeling strategies (Injection Timing and Network Security Tools), offering different perspectives on malicious activities.
[f] ICS-ADD is provided in PCAP format, lacking predefined features and attack percentage annotations, requiring preprocessing and manual labeling for supervised learning.

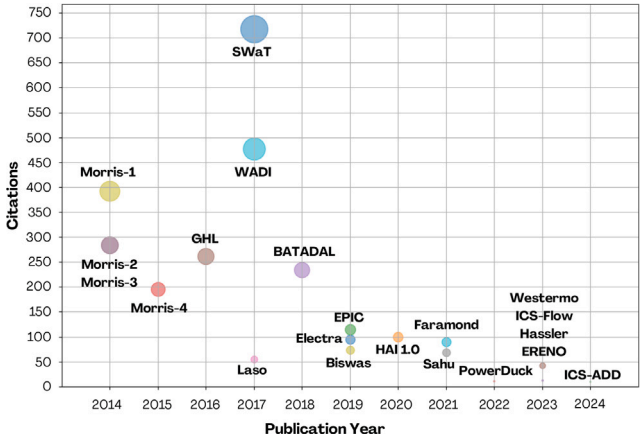


Fig. 2. Datasets citations. Retrieved on February 15, 2025.

that IDSs designed with unified datasets remain robust and applicable in real-world environments.

Finally, Fig. 2 shows the number of citations (size of the circles) of datasets and the year of their publication, based on Google Scholar data from February 2025. The SWaT dataset stands out as the most cited, with 724 citations. Below it, we observe various datasets such as Morris-1, Morris-2, WADI, and GHL, each with a substantial number of citations, suggesting these datasets have been significant for research in the field of CPS. Also, it can be seen that datasets published more recently, like ERENO, Westermo, and ICS-Flow, still have few citations, given the limited time they have had to be disseminated and utilized.

4.2. Source and features

Cyber features represent network and system-level observations, such as latency, throughput, or packet loss, which reflect the state of the cyber infrastructure. Physical features, on the other hand, correspond to measurements from the physical components of CPSs, such as the status of a circuit breaker (open/closed) or temperature readings from sensors. Both types of features serve as inputs to machine learning algorithms for detecting deviations or identifying patterns associated with attacks.

Two of the main sources of features for detecting intrusions in CPSs are application logs and network traffic. Application logs generate *host-based* features (Martins et al., 2022), while network traffic originates the so-called *network-based* features, two types of a more general class of *cyber* features.

However, the physical processes controlled by the CPS application serve as an additional valuable data source that can provide insights that extend beyond software interactions and are also amenable to analysis by an IDS. Unlike conventional application logs, which predominantly capture software interactions, data extracted from the physical features of a CPS offer a deeper understanding of the system's real-world behavior.

When physical features are considered, it is harder for an attacker to mask their actions because they will reflect in the observed physical behavior. Therefore, even if an attacker could trick the system and cover up its actions to be undetected at the cyber level, the physical features may still have information that may reveal an unexpected state (*e.g.*, a power transmission line being energized during maintenance).

Thus, by harnessing the synergistic potential of both cyber and physical features, IDSs in CPSs not only enhance their capability to detect anomalies but also ensure a comprehensive security that effectively bridges the gap between digital commands and physical outcomes.

Table 4
Physical components used to extract Physical Features, where M: Monitoring, C: Control, P: Processing, D&T: Distribution and Transmission, G: Generation, PR: Protection, ME: Measurement, CB: Circuit Breakers, HPN: Hermetic Pipe Networks, PG: Power Generators, PLC: Programmable Logic Controller, HMI: Human–Machine Interface.

System	Dataset	Physical components	Function
Water systems	SWaT	Tanks, Valves, Sensors, Pumps, Actuators	M, C, P
	WADI	Tanks, Valves, Sensors, Pumps, Reservoirs	D&T, M
	BATADAL	Storage Tanks, Sensors, Valves, Pumps, Reservoirs, Pipes, Junctions	D&T, M
	Faramond	Tanks, Valves, Sensors, Pumps, Reservoirs, Pipes	D&T, M
	Morris-3	Storage Tanks, Valves, Pumps, Sensors	M, C
	Laso	Tanks, Sensors, Pumps	M, C
UAV	Hassler	DJI Tello EDU, Barometer, Gyroscope, Accelerometer, Battery	M, C
Gas pipelines	Morris-2	HPN, Compressors, Pressure Gauges, Valves	M, C, D&T
	Morris-4	HPN, Compressors, Pressure Gauges, Valves	M, C, D&T
	GHL	Tanks, Pipe, Valves, Pumps, Sensors	M, C
Smart grids	Morris-1	CB, Relays, PG, IEDs, PMUs	G, M, C, D&T
	ERENO	CB, Transformers, IEDs, MU	M, C, D&T
	EPIC	CB, Motor-driven Generators, Photovoltaic Panels, Battery Systems, IEDs, PLCs, Inverters, Load Banks	G, D&T, ME, PR, C
	Biswas	CB, Relays, PG, Transformers, IEDs, Buses	ME, PR, C
	PowerDuck	CB, Station Control System	ME, PR, C
	Sahu	PMUs, Smart Meters	M, C
ICS	Westermo	PLCs, HMI, Industrial Routers	M, C
	HAI 1.0	GE Turbine, Emerson Boiler, FESTO Water Treatment Systems	M, C
	ICS-ADD	PLCs, HMI	M, C

4.3. Physical components

Table 4 illustrates the physical components commonly used to extract physical features from the studied datasets. In CPSs, the composition of physical components is crucial for capturing accurate and representative data that can be used for monitoring, control, and other important systemic functions.

Examples can be found in different applications, such as water treatment and gas pipeline systems. In water treatment and distribution systems (system at the top of Table 4), it is typical to use physical components such as tanks, valves, sensors, pumps, and reservoirs (Goh et al., 2016; Faramondi et al., 2021; Taormina et al., 2018). These elements are essential to ensure that the monitoring and control process in water systems can occur efficiently and correctly. In gas pipeline systems, it is common to find components like hermetic pipeline networks, compressors, pressure gauges, storage tanks, and valves (Morris and Gao, 2014; Morris et al., 2015). Data collected from compressors, for example, may include temperature readings. Valves, on the other hand, are crucial for regulating the flow and pressure of gas. Therefore, all of these components are fundamental for safe monitoring, control, distribution, and transportation of gas.

The most common components of smart grids are Circuit Breakers (CB), relays, transformers, IEDs, and Merging Units (MU). These elements enable effective monitoring and control of the generation, distribution, and transmission of electrical energy. For example, circuit breakers and relays provide physical data on power interruptions and restorations, essential for features that help understand load patterns and respond to network failures.

It is important to note that, while there are components frequently used in certain types of infrastructure, the variation among datasets is significant, reflecting the unique architecture of each system. For example, the EPIC dataset, in the context of smart grids, stands out for the diversity of its physical components. In addition to the most common elements mentioned above (*e.g.*, CB, IEDs), it also includes photovoltaic panels, battery storage systems, and engine-driven generators, demonstrating the complexity and multifunctionality required by modern grid systems (Adepu et al., 2019).

In addition to these infrastructure-based CPSs, UAVs introduce a distinct set of physical components, as they rely on onboard sensors and actuators for navigation and stability. While UAVs are not traditionally considered part of critical infrastructure like water or energy systems, they play an increasingly important role in applications such as emergency response, infrastructure inspection, and security operations. As seen in datasets like Hassler et al. (2023), drones rely on real-time sensor data from barometers, accelerometers, and gyroscopes for stable navigation.

This architectural heterogeneity across domains also affects feature relevance. Process measurements and actuator states may be particularly informative in water storage and gas pipeline scenarios (Gulzar and Mustafa, 2024; Transportation Security Administration, 2018), while in UAV datasets, the utility of features may depend more strongly on telemetry, onboard sensing, and communication status (Mykytyn et al., 2024). Accordingly, feature sets that are effective in one CPS domain should not be assumed to transfer uniformly to another.

In summary, Table 4 serves as a representation of the variety of physical components employed in CPSs, highlighting the need to understand each dataset within its specific operational context to improve the safety of these systems.

To provide a consolidated view of the datasets analyzed, we present a synthesized categorization in Table 5, summarizing the main characteristics, including application domains, attack types, number of samples, features, and data sources. This table allows direct comparison and highlights the diversity of datasets used in IDS research for CPSs.

4.4. Opportunities

The content and structure of data in CPS networks and devices vary according to the domain and application. The physical features are

Table 5
Synthesized categorization of CPS intrusion datasets.

Application	Dataset	Source	Feature type	Number samples	Number features	Number attacks	Attack type	Attack percentage	Data source
Water treatment	SWaT (Goh et al., 2016)	Sensors and Actuators, Network Sniffer	Cyber, Physical	946,722	51	1	Injection	5.87%	Testbed
Water distribution	WADI (Ahmed et al., 2017)	Sensors and Actuators	Cyber, Physical	172,803	130	15	Spoofing	5.77%	Testbed
	BATADAL (Taormina et al., 2018)	Sensors and Actuators	Cyber, Physical	12,940	45	14	Replay	1.69%	Simulated
	Faramond (Faramondi et al., 2021)	Sensors and Actuators, Network Sniffer	Cyber, Physical	24,306,718	56	4	DoS, MitM, Scanning, Physical	68.08%	Testbed
Water storage	Morris-3 (Morris and Gao, 2014)	Network Sniffer, Application Logs	Cyber, Physical	236,179	23	7	Injection, Reconnaissance, DoS	27.00%	Testbed
Water/Fuel Storage & Distribution	Laso (Laso et al., 2017)	Sensors, Network Sniffer	Cyber, Physical	337,438	3	3	DoS, Spoofing, Physical	30.98%	Testbed
UAV	Hassler (Hassler et al., 2023)	Sensors and Actuators, Network Sniffer, Application Logs	Cyber, Physical	54,784	53	4	De-Authentication (DoS), Replay, Injection, Evil Twin	74.97%	Testbed
Gas pipeline	Morris-2 (Morris and Gao, 2014)	Network Sniffer, Application Logs	Cyber, Physical	97,019	26	7	Injection, Reconnaissance, DoS	36.96%	Testbed
	Morris-4 (Morris et al., 2015)	Network Sniffer, Application Logs	Cyber, Physical	274,628	16	35	Injection, Reconnaissance, DoS	21.87%	Testbed
	GHL (Filonov et al., 2016)	Sensors and Actuators	Cyber, Physical	11,356,701	23	4	Control/Command Manipulation	–	Simulated
Smart grids	Morris-1 (Hink et al., 2014)	Network Sniffer, Application Logs	Cyber, Physical	76,035	128	40	Injection, Relay Setting Change	77.85%	Testbed
	ERENO (S.E. Quincozes et al., 2024)	Network Sniffer	Cyber, Physical	5,911,296	69	7	Injection, Masquerade, Replay, DoS	7.28%	Simulated
	EPIC (Adepu et al., 2019)	Sensors and Actuators	Cyber, Physical	5230	292–391	3	MitM, Malware Attack, Physical Damage	–	Testbed
	Biswas (Biswas et al., 2019)	Network Sniffer, Application Logs	Cyber, Physical	344,858	PCAP	3	DoS, Message Suppression, Data and Control Manipulation	89.77%	Simulated
	PowerDuck (Zemanek et al., 2022)	Network Sniffer, Application Logs	Cyber, Physical	1,133,589	Varies	4	Replay, Insertion, Suppression, Flooding	97.62%	Testbed
	Sahu (Sahu et al., 2021)	Network Sniffer, Application Logs	Cyber, Physical	–	28	3	Injection, MitM, False Data	–	Simulated
Industrial control system	ICS-Flow (Dehlaghi-Ghadim et al., 2023)	Network Sniffer, Application Logs	Cyber	45,719	64	4	Reconnaissance, Replay, Distributed DoS, MitM	33.86% & 19.71%	Simulated
	Westermo (Strandberg et al., 2023)	Network Sniffer, Application Logs	Cyber, Physical	68,730	64	6	Harmless/Bad SSH, Misconfigured/Duplicated IP Address, Port-Scan, MitM	21.39% & 65.30%	Testbed
	Electra (Gómez et al., 2019)	Network Sniffer, Application Logs	Cyber	16,289,279	11	3	Reconnaissance, Injection, Replay	4.42%	Testbed
	HAI 1.0 (Shin et al., 2020)	Sensors and Actuators	Cyber, Physical	995,404	64	3	Control Manipulation, Stealthy Attacks	1.83%	Testbed
	ICS-ADD (Gaggero et al., 2024)	Network Sniffer, Application Logs	Cyber, Physical	215,456	–	7	Reconnaissance, MitM, Injection, Brute-force, DNS Tunneling, DoS	–	Testbed

directly related to the domain (*e.g.*, current and voltage are common for smart grids but not for water treatment systems), while the cyber features depend on the specific communication protocols used for each domain. Therefore, building a general dataset that represents multiple domains of CPS may not be feasible. This challenge offers a unique opportunity to develop targeted, domain-specific datasets that cater to the nuances of each application.

Moreover, even though CPS datasets are available for some domains, recent works often continue to rely on traditional IT datasets, primarily network traffic data, to simulate attacks (Yang et al., 2019), indicating a slow integration of these specialized datasets. This reliance fails to capture the complex traffic patterns and interactions between physical and cyber components in CPSs, reducing the representativeness of CPS behaviors and attack scenarios and impacting the effectiveness of intrusion detection models. Also, to the best of our knowledge, important domains (e.g., transportation and chemical) lack intrusion detection datasets, providing a significant opportunity for research and development.

As utilities and private companies that manage real applications typically do not share their data, creating physical and virtual testbeds can be an alternative for researchers to acquire datasets from CPSs (Tushkanova et al., 2023; Umer et al., 2022). Whereas physical testbeds can produce more realistic messages due to the use of real equipment, the high cost of this approach may become a limitation to the construction of realistic scenarios. Additionally, reproducing attacks in physical production systems is often restricted by operators to protect sensitive and expensive equipment; for example, a relay setting change attack in power grid systems, which could alter protection configurations, might be avoided to prevent potential damage to relays or the need for costly recalibrations. Moreover, reproducing attacks in physical or virtual testbeds may be challenging due to the inherent complexities and variables involved. Using traffic generation tools is an opportunity to overcome this issue because such tools already enable non-expert users to use standard and commonly used configurations.

Besides being realistic and contemplating up-to-date attack models, to enable accurate intrusion detection, the data must be representative, *i.e.*, contain features that are non-redundant and descriptive of potential attacks and benign traffic. Traffic volume statistics, for example, may not be strongly related to an injection attack. Similarly, using physical information such as current and voltage measurements or the water pump state may not be enough to detect attacks like masquerade attacks. To address the information representativeness-related challenges, two preprocessing methods can improve the results of IDSs, namely *feature selection* and *feature enrichment*.

Feature selection aims to improve dataset quality by removing irrelevant, redundant, or less informative features that do not significantly contribute to attack detection. By selecting the most relevant attributes, this technique enhances model efficiency and reduces computational cost, leading to improved IDS performance.

Aside from feature selection, it may be necessary to enrich the available information to produce more representative features from the existing ones. The feature enrichment process may explore several techniques, such as (i) cross-layer feature correlation (*e.g.*, cyber and physical) (Quincozes et al., 2024), (ii) feature merging or derivation (*e.g.*, speed is derived from distance and time), (iii) event rate, frequency, or period, and (iv) time consistency between sequential messages (S.E. Quincozes et al., 2024; Skopik et al., 2020).

While feature selection and enrichment are established methods for enhancing ML models, their application in the context of CPS is still in the early stages of investigation. This presents an opportunity for further research to provide a deeper understanding of how these methods can be tailored to meet the specific requirements of CPS.

Examining the datasets in Table 3, we see that there is an imbalance between attack samples and normal samples. In particular, for half

of the datasets, attacks comprise less than 25% of the samples and for some the proportion is even more extreme; for example for the BATADAL dataset, there are 14 different types of attacks and they account for only 1.69% of the samples. While those small proportions of attack samples may be representative of the reality given that attacks are expected to be sporadic, unbalanced datasets can induce a bias in the intrusion detection model, which may classify attacks as normal behavior. Such a strategy can increase overall reported accuracy (most of the samples are indeed normal!), but it also risks elevating the number of undetected attacks, especially when the model lacks sufficient capability to identify the attacks.

Addressing the issue of unbalanced datasets in CPS is another opportunity, and we suggest the adoption of two strategies from ML in general, namely (a) changing the metric to be observed and (b) making the datasets more balanced. To overcome the inherent limitations of unbalanced datasets and to avoid over-reliance on accuracy — a metric that may not adequately reflect the system's performance in conditions of class imbalance — we advocate for the use of the F1-Score as a more comprehensive performance metric. The F1-Score considers both precision (the number of true positive results divided by the sum of true positives and false positives) and recall (the number of true positives divided by the sum of true positives and false negatives). This dual consideration encapsulates a more nuanced understanding of the model's performance, allowing for a clearer assessment of its effectiveness in distinguishing between legitimate and malicious activities in CPS. Additionally, minimizing the number of false alarms is crucial, as excessive false positives can compromise the credibility of the IDS (Quincozes et al., 2021; Talukder et al., 2024).

To address the imbalance in CPS datasets, we suggest data augmentation. This technique can artificially increase the number of attack samples by generating realistic and varied instances, thereby enhancing the model's ability to learn and detect diverse types of attacks, and potentially reducing the risk of overlooking subtle or rare attack patterns (Khan et al., 2023; Talukder et al., 2024; Zhang and Liu, 2022). One approach to improving data augmentation for CPS datasets is the use of Generative Adversarial Networks (GANs), which can generate synthetic malicious traffic samples that mimic real attack patterns, increasing dataset diversity while preserving realistic characteristics. Unlike traditional oversampling, GANs introduce controlled variations that enhance IDS generalization without distorting attack signatures (Dunmore et al., 2023). Alternatively, under-sampling can be applied to reduce the dominance of the majority class, potentially mitigating model bias toward normal traffic. While this approach decreases the number of samples, it can help the IDS focus on attack patterns by preventing an overwhelming preference for benign instances. However, its effectiveness depends on maintaining a sufficient number of representative samples to preserve essential traffic characteristics and avoid information loss (Arifin et al., 2024).

Another opportunity identified in the datasets analyzed in Section 4 would be the addition of more complex attack scenarios. While traditional attacks such as DoS and Injection are well represented, more sophisticated threats remain underexplored. In particular, datasets rarely include multi-stage attacks, where adversaries execute multiple phases before fully compromising a system (Al Homoud et al., 2023; Xiao et al., 2024), or adversarial attacks against ML-based IDSs, where malicious actors craft attack samples specifically designed to evade detection (Apruzzese et al., 2022). Similarly, coordinated (Varshini and Latha, 2024) and distributed attacks (Shafi et al., 2024), where multiple system components are targeted simultaneously (Lu et al., 2024), are underrepresented. Expanding datasets to include such attack scenarios could significantly improve the ability of IDS models to detect real-world threats.

Additionally, our analysis reveals that many CPS datasets could be enhanced with more comprehensive metadata and standardized documentation. Some datasets provide minimal descriptions of their structure, attack scenarios, and collection environments, making it difficult for researchers to effectively utilize and compare them. Establishing a standardized metadata framework for CPS datasets would enhance reproducibility and facilitate better dataset utilization across different studies. This framework should explicitly define key aspects, including the data collection methodology (testbed, simulation, real-world capture), attack descriptions (e.g., vectors used, impact on the system), dataset structure (e.g., included features, collection frequency), and clear labeling of normal and attack traffic. Another opportunity is the large-scale integration of heterogeneous CPS datasets through consistent attack labeling and compatible cyber and physical feature representations. Recent work on cross-dataset label harmonization for IoT intrusion detection provides a starting point for this direction (Rodríguez et al., 2026).

A broader opportunity for future research lies in bridging the gap between dataset-driven ML evaluations and real-world operational deployment. Translating high-performing models from research to practice introduces challenges that current datasets often do not capture. For instance, real-world CPSs evolve over time due to sensor degradation, firmware updates, and maintenance cycles. Moreover, datasets generated via simulation often lack the natural statistical variation, transient sensor reading errors, and physical noise inherent to industrial environments, leading to models that expect overly clean baseline data. Consequently, models trained exclusively on static or idealized dataset snapshots may be more susceptible to concept drift and elevated false-positive rates in production.

Addressing these challenges may require continuous learning strategies and noise-resilient models. Additionally, operational deployment must account for strict hardware constraints. While more complex models may achieve very high metrics on datasets using high-performance laboratory hardware, they may exceed the computational and memory limits of legacy field devices, such as PLCs or gateways. Future dataset collection and IDS evaluation frameworks should therefore better incorporate operational constraints and natural system noise, encouraging the development of models that balance detection performance with resource efficiency and adaptability over time.

5. Experiments

To demonstrate the practical applicability of the proposed taxonomy, we conducted a series of experiments using publicly available datasets for various CPS applications. Specifically, we aim to assess whether the taxonomy effectively categorizes datasets in a way that highlights their strengths and limitations for intrusion detection. Our experiments evaluate how dataset characteristics — such as feature diversity, attack representation, and labeling strategies — impact IDS performance.

By analyzing the effectiveness of different datasets within the proposed taxonomy framework, we identify potential gaps and challenges in existing datasets, providing insights for future dataset design and improvement.

5.1. Methodology and evaluation framework

The experiments were performed using the scikit-learn library,[1] employing three intrusion detection machine learning models: Decision Tree (DT), Random Forest (RF), and XGBoost. The 7 datasets used in this study correspond to different CPS domains, as detailed in Table 6, and include applications such as water distribution, water storage, smart grid, and gas pipelines. Each dataset was preprocessed to ensure consistency, with categorical columns encoded using label encoding. For datasets that did not provide predefined training and testing splits, we used scikit-learn's 80%/20% division; however, for datasets that specify their own partitioning (e.g., ERENO, which allocates 50% for

[1] https://scikit-learn.org/.

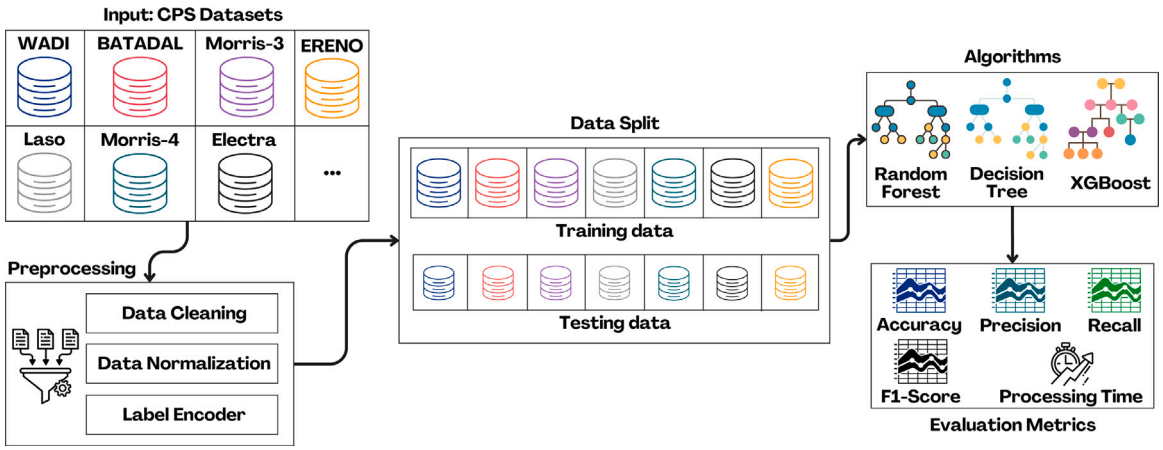


Fig. 3. Overview of the evaluation framework for intrusion detection datasets in cyber–physical systems (CPS). The methodology follows a structured approach, starting with dataset selection (BATADAL, Morris-3, Laso, Morris-4, Electra, WADI, ERENO), followed by data preprocessing, including cleaning, normalization, and label encoding. The data is then split into training and testing sets before applying machine learning models (Random Forest, Decision Tree, XGBoost) for intrusion detection. Finally, model performance is assessed using accuracy, precision, recall, F1-Score, and processing time metrics.

training and 50% for testing), we adhered to the original distribution to maintain consistency. Each model was evaluated using four performance metrics: Accuracy, Precision, Recall, and F1-Score. For Precision, Recall, and F1-Score, both *macro* and *weighted* versions were computed. To formally define these, let M_i represent the metric score for class i, n the total number of classes, S_i the number of samples (support) for class i, and S_{total} the total number of samples in the dataset.

Macro metrics treat all classes equally by computing the unweighted arithmetic mean, as defined in Eq. (1):

$$\text{Macro} = \frac{M_1 + M_2 + \cdots + M_n}{n} \tag{1}$$

In contrast, weighted metrics compute the average by weighting the score of each class by its support, accounting for the specific class distribution (Eq. (2)):

$$\text{Weighted} = \frac{M_1 S_1 + M_2 S_2 + \cdots + M_n S_n}{S_{total}} \tag{2}$$

In the context of imbalanced CPS datasets, macro metrics are generally prioritized to help ensure that the detection performance of minority classes is not masked by a predominant majority class. Aside from performance, the training and testing *processing times* were recorded to evaluate the computational efficiency of each approach.

The complete evaluation process is illustrated in Fig. 3. To ensure reproducibility and enable fair comparisons across different studies, we have formalized this pipeline into a Standardized Evaluation Framework (SEF-CPS) and made its codebase publicly available.[2] This framework automates the entire evaluation lifecycle — from dataset-specific data ingestion to the generation of consolidated reports containing classification metrics, per-instance inference times, and model structural complexity.

5.2. Results and discussion

Table 6 summarizes the experimental results, showing the performance of each model across the selected datasets. Key observations are as follows.

Datasets such as WADI and Morris-3 achieved near-perfect results across all metrics, highlighting the richness of their feature sets, which include both cyber and physical attributes, as categorized in the proposed taxonomy. This diversity of features enables the models to detect a wide range of attack patterns, making them robust in various scenarios. Furthermore, the physical nature of these specific domains inherently facilitates detection for tree-based algorithms. Water systems (WADI) operate within strict, slow-moving physical thresholds (e.g., tank levels and pipe pressure limits), which are naturally well-captured by the hierarchical decision boundaries of Random Forest and XGBoost. Conversely, electrical domains like smart grids exhibit rapid, complex transients (e.g., legitimate current spikes during equipment startup), making malicious deviations much harder to isolate using simple thresholds without advanced cross-layer feature enrichment.

On the other hand, Electra also exhibited strong classification performance, not necessarily because of its lack of physical features, but rather due to the nature of its cyber-only attacks being more distinguishable from normal traffic. However, this also indicates a limitation in generalizing to real-world scenarios where cyber–physical correlations are crucial to detect more complex interactions and multi-stage attacks, ultimately increasing robustness against a more diverse range of threats.

Although some datasets yielded near-perfect performance, this should not be interpreted as evidence that the evaluation is trivial. Rather, it suggests that certain datasets provide more clearly separable feature spaces, either because they combine informative cyber and physical attributes or because their attack patterns are more distinct from normal behavior. In this context, the strong performance of the DT model is particularly informative; as an automated rule-inducing algorithm, the DT indicates that part of this separability can be captured through relatively simple and interpretable decision structures. At the same time, this behavior is not uniform across all datasets, which reinforces the usefulness of the proposed taxonomy for explaining how characteristics such as feature richness, attack representation, and class distribution influence IDS performance.

While WADI and Morris-3 exhibited high classification performance with efficient training times, the Electra Modbus dataset required significantly more time for both training and inference, particularly with the Random Forest model. This behavior was expected, as Electra contains 16,289,279 samples, whereas Morris-3 and WADI have only 1.45% and 1.06% of this amount, respectively. The considerably larger dataset size naturally increases computational demands. However,

[2] The SEF-CPS framework, including scripts for automated data ingestion, training, and report generation, is available at: https://github.com/vagnerereno/sef-cps.

Table 6
Results of evaluation of the richness of different datasets from various applications, considering different feature types, data collection sources, and attack scenarios, based on the proposed taxonomy and performance metrics.

Application	Dataset	Model	Accuracy	Precision		Recall		F1-Score		Train time (s)	Test time (s)
				Macro	Weighted	Macro	Weighted	Macro	Weighted		
Water distribution	WADI	RF	1.00	1.00	1.00	1.00	1.00	1.00	1.00	33.43	0.19
		DT	1.00	1.00	1.00	1.00	1.00	1.00	1.00	7.18	0.01
		XGBoost	1.00	1.00	1.00	1.00	1.00	1.00	1.00	3.04	0.01
Water distribution	BATADAL	RF	0.82	0.87	0.84	0.54	0.82	0.53	0.75	6.63	0.03
		DT	0.80	0.63	0.75	0.56	0.80	0.56	0.76	0.48	0.003
		XGBoost	0.81	0.82	0.82	0.53	0.81	0.51	0.74	0.38	0.013
Water storage	Morris-3	RF	0.99	0.98	0.99	0.98	0.99	0.98	0.99	16.07	0.46
		DT	0.99	0.98	0.99	0.98	0.99	0.98	0.99	0.36	0.01
		XGBoost	0.96	0.94	0.97	0.98	0.96	0.95	0.97	3.84	0.06
Water/Fuel Storage & Distribution	Laso	RF	0.69	0.66	0.60	0.48	0.69	0.48	0.58	11.61	0.46
		DT	0.69	0.65	0.60	0.48	0.69	0.48	0.58	0.28	0.01
		XGBoost	0.69	0.74	0.69	0.48	0.69	0.46	0.57	1.96	0.03
Gas pipeline	Morris-4	RF	0.91	0.90	0.91	0.70	0.91	0.76	0.90	15.42	0.73
		DT	0.91	0.90	0.91	0.71	0.91	0.77	0.90	0.59	0.01
		XGBoost	0.90	0.91	0.90	0.67	0.90	0.75	0.89	4.53	0.07
Smart grid	ERENO	RF	0.97	0.75	0.98	0.74	0.97	0.72	0.99	155.6	22.38
		DT	0.96	0.69	0.97	0.73	0.96	0.68	0.97	6.04	0.34
		XGBoost	0.98	0.81	0.98	0.76	0.98	0.78	0.98	64.02	2.248
Industrial control system	Electra Modbus	RF	1.00	0.99	1.00	0.98	1.00	0.98	1.00	622.9	17.92
		DT	1.00	0.99	1.00	0.98	1.00	0.99	1.00	12.01	0.35
		XGBoost	1.00	1.00	1.00	0.98	1.00	0.99	1.00	322.37	1.84

Table 7
Average test time per instance (in microseconds) for each model and dataset.

Dataset	Random Forest	Decision Tree	XGBoost
WADI	5.50 µs	0.29 µs	0.29 µs
BATADAL	15.90 µs	1.41 µs	6.18 µs
Morris-3	9.74 µs	0.21 µs	1.27 µs
Laso	5.92 µs	0.15 µs	0.44 µs
Morris-4	13.30 µs	0.18 µs	1.27 µs
ERENO	7.57 µs	0.12 µs	0.76 µs
Electra	5.50 µs	0.11 µs	0.56 µs

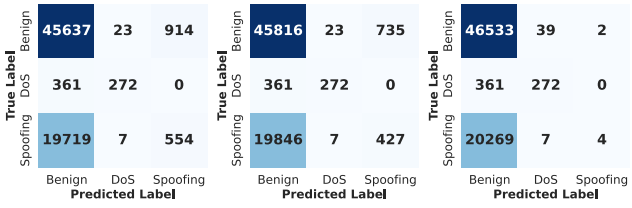


Fig. 4. Confusion matrices for the classification models applied to the Laso dataset: Random Forest (left), Decision Tree (center), and XGBoost (right). Each matrix shows the performance in distinguishing between classes "Benign" (normal), "DoS", and "Spoofing".

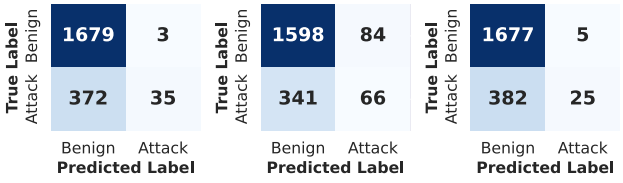


Fig. 5. Confusion matrices for the classification models applied to the BATADAL dataset: Random Forest (left), Decision Tree (center), and XGBoost (right). Values represent the number of samples classified as "Benign" (normal) or "Attack" (intrusion).

when examining the average test time per instance (Table 7), Electra and ERENO exhibit significantly lower times compared to smaller datasets like BATADAL. This indicates that, despite the large volume of data, the algorithms maintain per-instance efficiency, likely due to the structured nature of the data and the optimization characteristics of the models used. Additionally, an exploratory analysis facilitated by the framework's structural reporting indicated a possible relationship between tree depth and inference time; this analysis is not the focus of this paper and requires further investigation.

The Laso dataset, containing only three features, made it significantly harder for all models to detect the two attacks. This aligns with the taxonomy's emphasis on the importance of comprehensive feature representation for effective intrusion detection. As observed in the confusion matrices (Fig. 4), the models exhibited a notable tendency to misclassify "Spoofing" samples as "Benign", likely due to the limited feature set failing to capture distinguishing characteristics between these classes. The dataset's composition, with 337,438 samples and a relatively high attack percentage of 30.98%, further underscores the challenge: despite the balanced attack ratio, the restricted feature space hindered the models' ability to differentiate between attack types effectively. These limitations highlight the need for feature enrichment or alternative approaches to compensate for the restricted feature set, as discussed in Sections 3 and 4.

Datasets such as BATADAL, which have a low proportion of attack samples (1.69%), showed reduced performance in Recall and F1-Score for all models, despite achieving moderate Accuracy. This highlights a challenge identified in the taxonomy: the imbalance between attack and normal samples can lead to models favoring the majority class, thus misclassifying rare attack instances. As observed in the confusion matrices (Fig. 5), the models exhibited significant difficulty in correctly identifying attack samples. Another factor that may have contributed to the lower performance is the relatively small size of the dataset (12,940 samples in total), which is further divided into training and testing subsets. This limited sample size may have constrained the models' ability to generalize effectively, particularly for the minority class.

The Morris-4 dataset has 30% fewer features than Morris-3 (16 compared to 23) and smaller proportion of attack samples (21.87% compared to 27.00%). These characteristics likely contributed to the lower

performance observed in Morris-4, as fewer features and a smaller representation of attacks can limit the models' ability to effectively capture and distinguish patterns. This is further reflected in the metrics, where a noticeable gap exists between the macro and weighted averages for Recall and F1-Score. For Morris-4, the macro Recall and F1-Score were around 70%, while the weighted metrics reached 91% and 90%, respectively. This difference highlights the influence of the "Benign" class, which dominates the dataset and skews the weighted averages upward. Meanwhile, the macro metrics, treating all classes equally, expose the models' difficulties in detecting underrepresented or more complex attack classes, underscoring the challenges posed by class imbalance and feature limitations.

To conclude this analysis, we evaluated the ERENO dataset using only its basic set of 22 features — while the dataset contains a total of 69 features, 47 of those are enriched. The results indicate that while ERENO achieves a high accuracy, the macro F1-Score and other macro-averaged metrics are notably lower, suggesting classification errors among specific attack types. This likely stems from the model struggling to correctly classify certain attack classes. These findings highlight the challenges of working with a reduced feature set in complex CPS environments. It is worth noting that previous work by the authors of the ERENO dataset has demonstrated how feature selection and enrichment can improve detection performance; interested readers can refer to S.E. Quincozes et al. (2024) for further details.

A limitation of CPS security research is the restricted access to private industrial datasets due to several factors, including protection of intellectual property, trade secrets, and strict confidentiality constraints. Building upon the real-world operational challenges discussed in Section 4.4, private datasets may differ from public testbeds by involving proprietary or insufficiently documented protocols (Wressnegger et al., 2018) and by offering less reliable or less verifiable ground-truth attack labels (Landauer et al., 2022). In addition, private operational datasets may expose IDS models to previously unseen conditions, which is consistent with the broader IDS challenge of handling novel attacks and new, unseen data over time (Shyaa et al., 2024). Consequently, evaluating IDS models on private data may yield performance outcomes that differ from those observed in public testbeds — such as variations in detection rates or false positive levels — depending on the specific environment's complexity. Nevertheless, the methodology applied in this study remains highly relevant, as the SEF-CPS evaluation framework (Section 5.1) is designed to be dataset-agnostic. This enables owners of private industrial datasets to systematically evaluate their own defenses. Provided that the data can be structured with extractable features and labels, our framework offers a standardized pipeline to assess feature richness and IDS effectiveness, regardless of whether the data originates from a public testbed or a private industrial facility.

5.3. Feature contribution analysis

To further investigate how specific feature categories influence IDS performance and address the challenge of feature representativeness, we conducted an analysis across three datasets representing different applications: Morris-3 (Water Storage), ERENO (Smart Grid), and Morris-4 (Gas Pipeline). Table 8 details all the features present in these datasets, categorized into cyber and physical groups according to our proposed taxonomy, as well as their respective target classes (comprising both benign/normal operations and specific cyberattacks). To ensure fidelity and reproducibility, both feature names and class labels are maintained exactly as provided by the original dataset authors.

As discussed in Section 5.1, we prioritize the macro F1-Score to ensure that the detection performance of minority classes is not masked by the predominant majority class. Fig. 6 illustrates the macro F1-Score achieved by the Random Forest, Decision Tree, and XGBoost models when evaluated across different feature sets: exclusively cyber features, exclusively physical features, and the combined cyber–physical feature set.

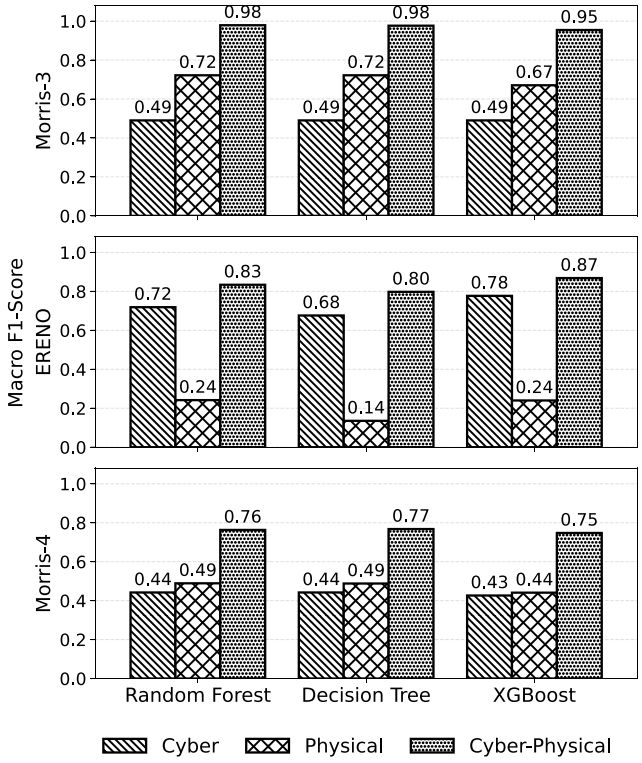


Fig. 6. Macro F1-Score comparison of Random Forest, Decision Tree, and XGBoost models trained on isolated (Cyber-only, Physical-only) versus integrated (Cyber–Physical) feature sets across the Morris-3, ERENO, and Morris-4 datasets.

As observed, models trained on isolated feature categories exhibit lower performance across the datasets. For instance, in Morris-3 and Morris-4, relying exclusively on cyber features resulted in macro F1-Scores below 0.50 across all tested algorithms. Conversely, when models were trained exclusively with physical features, performance was particularly low for the ERENO dataset, with macro F1-Scores remaining below 0.25 (while Morris-4 achieved scores around 0.50, and Morris-3 reached up to 0.72 in this scenario). However, the integration of both cyber and physical features consistently improved the results. When combined, the models achieved their highest macro F1-Scores, reaching maximum values of 0.98 in Morris-3, 0.87 in ERENO, and 0.77 in Morris-4 across the evaluated algorithms. This suggests that combining data from both domains provides a more comprehensive representation of the system state, aiding the models in differentiating between normal traffic and various attacks.

To understand the individual contribution of these features to the classification models, we applied SHapley Additive exPlanations (SHAP), an eXplainable Artificial Intelligence (XAI) technique (V.E. Quincozes et al., 2024). Figs. 7, 8, and 9 present the SHAP global importance for the top-15 features of the best-performing model in each dataset. To facilitate interpretation and align with our proposed taxonomy, we appended a '(C)' or '(P)' to each feature name in the plots, explicitly denoting its classification as a cyber or physical feature, respectively. Analyzing these top features across diverse scenarios reveals how models leverage both cyber and physical domains, as well as their specific relevance to different attack classes.

In the Morris-3 dataset (Fig. 7), the feature importance ranking is led by a mix of physical and cyber attributes. The physical *measurement* and the cyber *response_address* present the highest overall impact on the model's output. Notably, the multiclass SHAP analysis reveals distinct

Table 8
Categorization of all features (cyber and physical) and classes for Morris-3, ERENO, and Morris-4 datasets.

Dataset	Cyber Features	Physical Features	Classes
Morris-3	command_address, response_address, command_memory, response_memory, command_memory_count, response_ memory_count, comm_read_function, comm_write_fun, resp_read_fun, resp_ write_fun, sub_function, command_ length, resp_length, crc_rate	HH, H, L, LL, control_mode, control_scheme, pump, measurement	Benign, Reconnaissance (Recon), DoS, Malicious Parameter Command Injection (MPCI), Naive Malicious Response Injection (NMRI), Complex Malicious Response Injection (CMRI), Malicious State Command Injection (MSCI), Malicious Function Code Injection (MFCI)
ERENO	t, GooseTimestamp, SqNum, StNum, cbStatus, frameLen, ethDst, ethSrc, ethType, gooseTimeAllowedtoLive, gooseAppid, gooseLen, TPID, gocbRef, datSet, goID, test, confRev, ndsCom, numDatSetEntries, APDUSize, protocol	Time, isbA, isbB, isbC, ismA, ismB, ismC, vsbA, vsbB, vsbC, vsmA, vsmB, vsmC	Benign (Normal), Masquerade Fake Normal, Random Replay, Poisoned High Rate, Inverse Replay, Masquerade Fake Fault, Injection, High StNum
Morris-4	address, function, length, crc_rate, command_response	setpoint, gain, reset_rate, deadband, cycle_time, rate, system_mode, control_scheme, pump, solenoid, pressure_measurement	Benign, Recon, DoS, MPCI, NMRI, CMRI, MSCI, MFCI

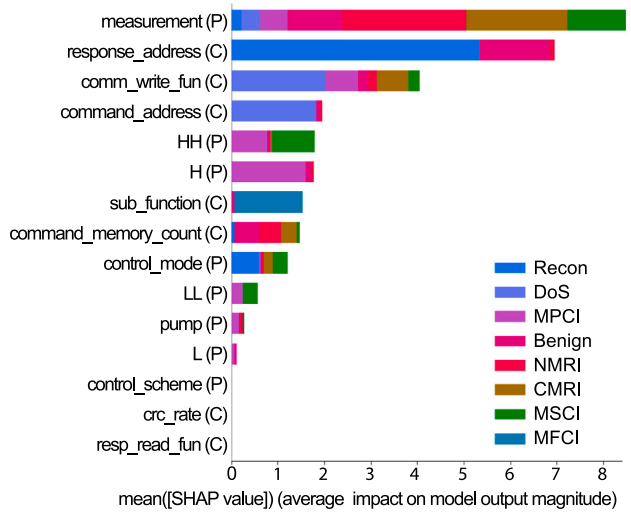


Fig. 7. SHAP global feature importance (top-15) for the Random Forest model evaluated on the Morris-3 dataset. This cyber–physical configuration yielded the highest classification performance, achieving a macro F1-Score of 0.98.

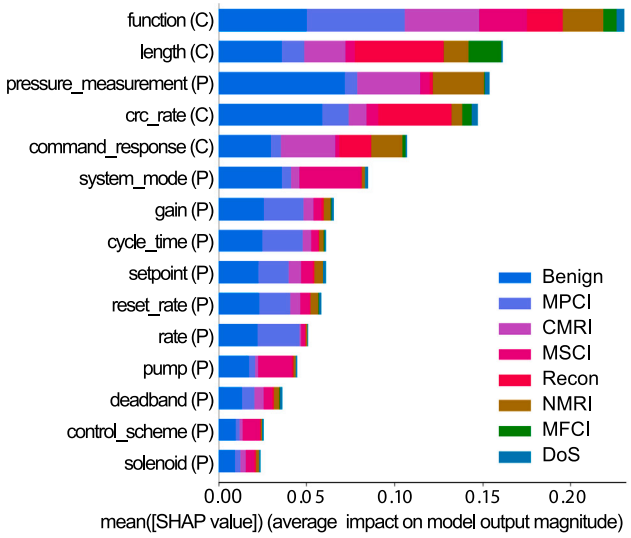


Fig. 8. SHAP global feature importance (top-15) for the Random Forest model evaluated on the Morris-4 dataset. The model leverages both network commands and thermodynamic process variables, achieving a macro F1-Score of 0.76.

specializations among these features. While *measurement* acts as a broad indicator that contributes to distinguishing among nearly all classes — showing a substantial impact on Naive Malicious Response Injection (NMRI) and Complex Malicious Response Injection (CMRI) attacks — the *response_address* feature is overwhelmingly important for identifying Reconnaissance activities. Other physical features, such as *H* (High setpoint) and *HH* (High alarm setpoint), also appear in the top 10, highlighting their relevance for identifying specific process anomalies.

A similar pattern is observed in the Morris-4 gas pipeline dataset (Fig. 8). While cyber protocol features like *function* and *length* occupy the top positions, they are immediately followed by *pressure_measurement* (physical). In this scenario, the SHAP plot demonstrates that these top features are highly effective at establishing the baseline for benign samples, while simultaneously providing the necessary variance to classify the different types of injection and denial-of-service attacks.

In the ERENO smart grid dataset (Fig. 9), cyber features related to the GOOSE protocol, such as *SqNum*, *frameLen*, and *cbStatus*, dominate the uppermost positions. For instance, *SqNum* exhibits a high contribution spread across almost all classes in the dataset, proving its versatility as a universal discriminator between normal and malicious traffic. This aligns with the highly deterministic nature of substation communication, where sequence numbers and status flags are critical indicators of attacks like masquerade or replay (S.E. Quincozes et al., 2024). However, physical attributes such as *Time*, *vsmA*, and *ismA* (voltage and current measurements) are firmly present within the top 15 features. Their presence confirms that while protocol syntax is crucial, the underlying electrical variables provide essential context for the IDS to confirm the legitimacy of the network events.

Overall, the SHAP analysis across all three datasets corroborates that effective intrusion detection in CPS environments benefits from a hybrid feature space. While network-level features often serve as primary indicators of malicious activity, physical process variables act as critical validators. These findings also suggest practical directions for mitigating dataset limitations: future CPS IDS datasets should preserve both cyber and physical features whenever possible, and feature refinement should be guided by domain-specific operational context and

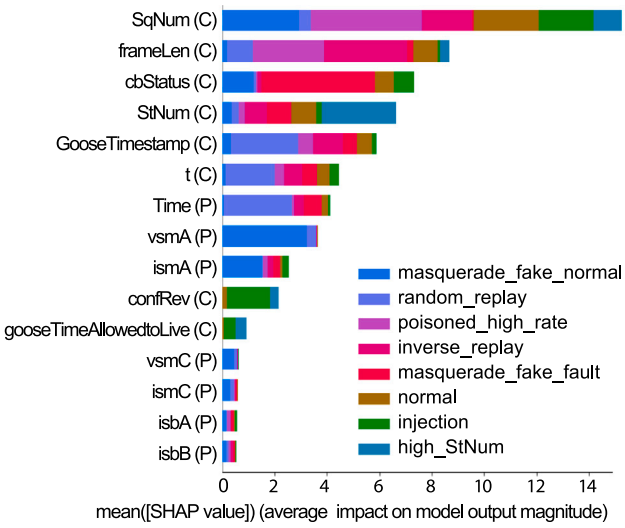


Fig. 9. SHAP global feature importance (top-15) for the XGBoost model evaluated on the ERENO dataset (identified as the best-performing model for this specific dataset). The synergistic use of IEC-61850 features and physical measurements resulted in a peak macro F1-Score of 0.87.

interpretability analyses to reduce representational gaps and improve robustness. A more detailed and exhaustive analysis of the individual impact of each feature on specific attack classes requires deeper operational context and is left for future work.

6. Conclusion

The prioritization of high availability and real-time responses over security requirements in CPSs, combined with increasing connectivity, has left critical infrastructure vulnerable. For this reason, it is extremely important to employ IDSs as a tool to detect attacks in such fundamental systems. However, ML-based IDSs need proper datasets containing representative attack signatures. While classical datasets are available for intrusion detection in IT systems, they do not capture the peculiarities of CPSs.

Motivated by that, we reviewed the state-of-the-art intrusion datasets available for CPSs and discussed the challenges and opportunities of the area. We observed that CPS datasets are often fit-for-purpose, as different domains are naturally heterogeneous — varying not only in their communication protocols but also in their timing requirements, process dynamics, and operational constraints — and many of them lack publicly available datasets. To address this gap, we propose a taxonomy that categorizes datasets, features, and attack types, facilitating a more structured approach to CPS security research and enhancing comparability across studies.

To validate the applicability of our taxonomy, we applied it to seven publicly available CPS datasets spanning different domains, including water distribution, smart grids, and industrial control systems. To ensure reproducibility and enable standardized comparisons, this evaluation was formalized and released as the open-source SEF-CPS framework. This analysis allowed us to systematically identify dataset limitations, such as unbalanced attack distributions and insufficient feature diversity, reinforcing the need for dataset refinement techniques.

Future work could focus on addressing the challenges posed by feature-limited datasets, such as Laso, and imbalanced datasets, such as BATADAL, through feature enrichment techniques and data augmentation strategies. Additionally, exploring lightweight models for real-time IDS deployment could help mitigate the computational challenges observed in certain datasets, such as BATADAL. It would also be valuable to evaluate alternative classification approaches beyond tree-based models and investigate XAI-guided feature selection techniques aimed at improving feature relevance and detection performance while making model decisions more interpretable in CPS contexts.

Additional research directions include investigating distributed and federated classifier-based IDSs in CPSs, particularly their ability to handle heterogeneous local datasets and resist poisoning of exchanged information (Giovanni et al., 2026; Shan et al., 2025). Large-scale integration of heterogeneous CPS datasets through consistent attack labeling and compatible cyber and physical feature representations also warrants further investigation.

CRediT authorship contribution statement

Vagner E. Quincozes: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Conceptualization. **Silvio E. Quincozes:** Writing – review & editing, Visualization, Validation, Methodology, Investigation, Conceptualization. **Célio Albuquerque:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Conceptualization. **Diego Passos:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Conceptualization. **Daniel Mossé:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was carried out with the support of CAPES, CNPq, FAPERJ, and the United States Department of Energy under Award Number DE-CR0000039.

Data availability

No data was used for the research described in the article.

References

Adepu, S., Kandasamy, N.K., Mathur, A., 2019. EPIC: An Electric Power Testbed for Research and Training in Cyber Physical Systems Security. In: Computer Security: SECPRE CyberICPS. Springer, pp. 37–52.

Ahmad, Z., Shahid Khan, A., Wai Shiang, C., Abdullah, J., Ahmad, F., 2021. Network intrusion detection system: A systematic study of machine learning and deep learning approaches. Trans. Emerg. Telecommun. Technol. 32 (1), e4150.

Ahmed, C.M., Kandasamy, N.K., 2021. A Comprehensive Dataset from a Smart Grid Testbed for Machine Learning based CPS Security Research. In: International Workshop on Cyber-Physical Security for Critical Infrastructures Protection. Springer, pp. 123–135.

Ahmed, C.M., Palleti, V.R., Mathur, A.P., 2017. WADI: a water distribution testbed for research in the design of secure cyber physical systems. In: Int. Workshop on Cyber-Physical Systems for Smart Water Networks. pp. 25–28.

Al-Ghuwairi, A.-R., Sharrab, Y., Al-Fraihat, D., AlElaimat, M., Alsarhan, A., Algarni, A., 2023. Intrusion detection in cloud computing based on time series anomalies utilizing machine learning. J. Cloud Comput. 12 (1), 127.

Al Homoud, L., Barpanda, N., Goulart, A., Davis, K., Rice, M., 2023. On Grid Resiliency: Cyber-Physical Detection Tool Evaluated in a Multi-Stage Attack Scenario. In: IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids. SmartGridComm, IEEE, pp. 1–7.

Alshaibi, A., Al-Ani, M., Al-Azzawi, A., Konev, A., Shelupanov, A., 2022. The comparison of cybersecurity datasets. Data 7 (2), 22.

Alwan, A.A., Ciupala, M.A., Brimicombe, A.J., Ghorashi, S.A., Baravalle, A., Falcarin, P., 2022. Data quality challenges in large-scale cyber-physical systems: A systematic review. Inf. Syst. 105, 101951.

Apruzzese, G., Andreolini, M., Ferretti, L., Marchetti, M., Colajanni, M., 2022. Modeling realistic adversarial attacks against network intrusion detection systems. Digit. Threat.: Res. Pract. (DTRAP) 3 (3), 1–19.

Arifin, M.A.S., Stiawan, D., Yudho Suprapto, B., Susanto, S., Salim, T., Idris, M.Y., Budiarto, R., 2024. Oversampling and undersampling for intrusion detection system in the supervisory control and data acquisition IEC 60870-5-104. IET Cyber Phys. Syst.: Theory Appl.

Ashraf, W., Masoodi, F.S., 2026. A survey of intrusion detection datasets for communication networks. Discov. Comput. 29 (1), 577.

Biswas, P.P., Tan, H.C., Zhu, Q., Li, Y., Mashima, D., Chen, B., 2019. A synthesized dataset for cybersecurity study of IEC 61850 based substation. In: International Conference on Communications, Control, and Computing Technologies for Smart Grids. IEEE, pp. 1–7.

Chui, K.T., Gupta, B.B., Liu, J., Arya, V., Nedjah, N., Almomani, A., Chaurasia, P., 2023. A survey of internet of things and cyber-physical systems: Standards, algorithms, applications, security, challenges, and future directions. Information 14 (7), 388.

Conti, M., Donadel, D., Turrin, F., 2021. A survey on industrial control system testbeds and datasets for security research. IEEE Commun. Surv. Tutor. 23 (4), 2248–2294.

Dehlaghi-Ghadim, A., Moghadam, M.H., Balador, A., Hansson, H., 2023. Anomaly detection dataset for industrial control systems. IEEE Access 11, 107982–107996.

Dunmore, A., Jang-Jaccard, J., Sabrina, F., Kwak, J., 2023. A comprehensive survey of generative adversarial networks (GANs) in cybersecurity intrusion detection. IEEE Access.

Faheem, M., Butt, R.A., 2022. Big datasets of optical-wireless cyber-physical systems for optimizing manufacturing services in the internet of things-enabled industry 4.0. Data Brief 42, 108026.

Falliere, N., O Murchu, L., Chien, E., 2011. W32.Stuxnet Dossier. Technical Report, Symantec Corporation, URL: https://nsarchive2.gwu.edu/NSAEBB/NSAEBB424/docs/Cyber-044.pdf. (Accessed 12 October 2026).

Faramondi, L., Flammini, F., Guarino, S., Setola, R., 2021. A hardware-in-the-loop water distribution testbed dataset for cyber-physical security testing. IEEE Access 9, 122385–122396.

Filonov, P., Lavrentyev, A., Vorontsov, A., 2016. Multivariate industrial time series with cyber-attack simulation: Fault detection using an LSTM-based predictive data model. arXiv preprint arXiv:1612.06676.

Gaggero, G.B., Armellin, A., Portomauro, G., Marchese, M., 2024. Industrial control system-anomaly detection dataset (ICS-ADD) for cyber-physical security monitoring in smart industry environments. IEEE Access.

Giovanni, H.d.L., Chessio, M.E.S., Quincozes, S.E., Mossé, D., Quincozes, V.E., Albuquerque, C., Passos, D., 2026. Ataques de envenenamento de rótulos contra a detecçao de zero-day em sistemas de detecçao de intrusao colaborativos. In: Simpósio Brasileiro de Redes de Computadores e Sistemas Distribuídos. SBC, pp. 295–308.

Goh, J., Adepu, S., Junejo, K.N., Mathur, A., 2016. A dataset to support research in the design of secure water treatment systems. In: International Conference on Critical Information Infrastructures Security. Springer, pp. 88–99.

Goldschmidt, P., Chudá, D., 2025. Network intrusion datasets: A survey, limitations, and recommendations. Comput. Secur. 156, 104510.

Gómez, Á.L.P., Maimó, L.F., Celdrán, A.H., Clemente, F.J.G., Sarmiento, C.C., Masa, C.J.D.C., Nistal, R.M., 2019. On the generation of anomaly detection datasets in industrial control systems. IEEE Access 7, 177460–177473.

Gulzar, Q., Mustafa, K., 2024. An analytical survey of cyber-physical systems in water treatment and distribution: Security challenges, intrusion detection, and future directions. Secur. Priv. 7 (6), e440.

Hassler, S.C., Mughal, U.A., Ismail, M., 2023. Cyber-physical intrusion detection system for unmanned aerial vehicles. IEEE Trans. Intell. Transp. Syst.

Hink, R.C.B., Beaver, J.M., Buckner, M.A., Morris, T., Adhikari, U., Pan, S., 2014. Machine learning for power system disturbance and cyber-attack discrimination. In: International Symposium on Resilient Control Systems. IEEE, pp. 1–8.

Hussain, F., Abbas, S.G., Shah, G.A., Pires, I.M., Fayyaz, U.U., Shahzad, F., Garcia, N.M., Zdravevski, E., 2021. A framework for malicious traffic detection in IoT healthcare environment. Sensors 21 (9), 3025.

Hutzelmann, T., Mauksch, D., Petrovska, A., Pretschner, A., 2023. Generation of Tailored and Confined Datasets for IDS Evaluation in Cyber-Physical Systems. IEEE Trans. Dependable Secur. Comput.

Jamal, A.A., Majid, A.-A.M., Konev, A., Kosachenko, T., Shelupanov, A., 2023. A review on security analysis of cyber physical systems using Machine learning. Mater. Today: Proc. 80, 2302–2306.

Ji, R., Padha, D., Singh, Y., Sharma, S., 2024. Review of intrusion detection system in cyber-physical system based networks: Characteristics, industrial protocols, attacks, data sets and challenges. Trans. Emerg. Telecommun. Technol. 35 (9), e5029.

Katulić, F., Sumina, D., Groš, S., Erceg, I., 2023. Protecting modbus/TCP-based industrial automation and control systems using message authentication codes. IEEE Access 11, 47007–47023.

Kayan, H., Nunes, M., Rana, O., Burnap, P., Perera, C., 2022. Cybersecurity of industrial cyber-physical systems: A review. ACM Comput. Surv. 54 (11s), 1–35.

Khalid, F., Rehman, S., Shafique, M., 2020. Overview of security for smart cyber-physical systems. Secur. Cyber Phys. Syst.: Vulnerability Impact 5–24.

Khan, A.A., Chaudhari, O., Chandra, R., 2023. A review of ensemble learning and data augmentation models for class imbalanced problems: Combination, implementation and evaluation. Expert Syst. Appl. 122778.

Landauer, M., Skopik, F., Frank, M., Hotwagner, W., Wurzenberger, M., Rauber, A., 2022. Maintainable log datasets for evaluation of intrusion detection systems. IEEE Trans. Dependable Secur. Comput. 20 (4), 3466–3482.

Laso, P.M., Brosset, D., Puentes, J., 2017. Dataset of anomalies and malicious acts in a cyber-physical subsystem. Data Brief 14, 186–191.

Lin, P.-Y., Tien, C.-W., Huang, T.-C., Tien, C.-W., 2021. ICPFuzzer: proprietary communication protocol fuzzing by using machine learning and feedback strategies. Cybersecurity 4, 1–15.

Lin, Q., Verwer, S., Kooij, R., Mathur, A., 2020. Using datasets from industrial control systems for cyber security research and education. In: International Conference on Critical Information Infrastructures Security. Springer, pp. 122–133.

Ltd, M.H.I., 2022. The world's most advanced tool for power systems EMT simulations. https://www.pscad.com/software/pscad/overview.

Lu, Q., Gao, Q., Li, J., Xie, X., Guo, W., Wang, J., 2024. Distributed cyber-physical intrusion detection using stacking learning for wide-area protection system. Comput. Commun. 215, 91–102.

Martins, I., Resende, J.S., Sousa, P.R., Silva, S., Antunes, L., Gama, J., 2022. Host-based IDS: A review and open issues of an anomaly detection system in IoT. Future Gener. Comput. Syst. 133, 95–113.

Morris, T., Gao, W., 2014. Industrial control system traffic data sets for intrusion detection research. In: International Conference on Critical Infrastructure Protection. Springer, pp. 65–78.

Morris, T.H., Thornton, Z., Turnipseed, I., 2015. Industrial control system simulation and data logging for intrusion detection system research. In: 7th Annual Southeastern Cyber Security Summit. pp. 3–4.

Mykytyn, P., Brzozowski, M., Dyka, Z., Langendoerfer, P., 2024. A survey on sensor- and communication-based issues of autonomous UAVs. Comput. Model. Eng. Sci. (CMES) 140 (2).

Olowononi, F.O., Rawat, D.B., Liu, C., 2021. Resilient machine learning for networked cyber physical systems: A survey for machine learning security to securing machine learning for CPS. IEEE Commun. Surv. Tutor. 23 (1), 524–552.

Premaratne, U.K., Samarabandu, J., Sidhu, T.S., Beresh, R., Tan, J.-C., 2010. An intrusion detection system for IEC61850 automated substations. IEEE Trans. Power Deliv. 25 (4), 2376–2383.

Qassim, Q.S., Ali, M.A., Tahir, N.M., 2023. Security analysis of DNP3 protocol in SCADA system. In: International Conference on Control System, Computing and Engineering. pp. 314–319.

Quincozes, S.E., Albuquerque, C., Passos, D., Mossé, D., 2021. A survey on intrusion detection and prevention systems in digital substations. Comput. Netw. 184, 107679.

Quincozes, S.E., Albuquerque, C., Passos, D., Mossé, D., 2024. ERENO: A framework for generating realistic IEC–61850 intrusion detection datasets for smart grids. IEEE Trans. Dependable Secur. Comput. 21 (4), 3851–3865.

Quincozes, V.E., Quincozes, S.E., Kazienko, J.F., Gama, S., Cheikhrouhou, O., Koubaa, A., 2024. A survey on IoT application layer protocols, security challenges, and the role of explainable AI in IoT (XAIoT). Int. J. Inf. Secur. 1–28.

Quincozes, V.E., Quincozes, S.E., Passos, D., Albuquerque, C., Mossé, D., 2024. Towards feature engineering for intrusion detection in IEC–61850 communication networks. Ann. Telecommun. 79 (7), 537–551.

Reddy, D.S., Kumar, K.A., 2026. A dataset-centric review of IoT and IIoT intrusion detection: realism, evaluation biases, and future research directions. Front. Big Data 9, 1878260.

Rocchetto, M., Tippenhauer, N.O., 2016. On attacker models and profiles for cyber-physical systems. In: 21st European Symposium on Research in Computer Security. Springer, pp. 427–449.

Rodríguez, M., Mehavilla, L., Alesanco, Á., García, J., 2026. A rule-based label harmonization framework for cross-dataset IoT intrusion detection. Internet Things 39, 102030.

Sahu, A., Mao, Z., Wlazlo, P., Huang, H., Davis, K., Goulart, A., Zonouz, S., 2021. Multi-source multi-domain data fusion for cyberattack detection in power systems. IEEE Access 9, 118–138.

Shafi, M., Lashkari, A.H., Rodriguez, V., Nevo, R., 2024. Toward generating a new cloud-based distributed denial of service (DDoS) dataset and cloud intrusion traffic characterization. Information 15 (4), 195.

Shan, Y., Zhao, J., Song, Y., Teng, H., Hou, W., Liu, Z., 2025. CoperFed: A covert personalized federated learning framework for industrial control systems intrusion detection. J. Ind. Inf. Integr. 48, 101004.

Sheikh, Z.A., Singh, Y., 2023. Review of cyber-physical system-based security datasets for learning-based intrusion detection systems. In: The International Conference on Recent Innovations in Computing. Springer, pp. 291–305.

Shin, H.-K., Lee, W., Yun, J.-H., Kim, H., 2020. HAI 1.0: HIL-based augmented ICS security dataset. In: Cyber Security Experimentation and Test Workshop.

Shyaa, M.A., Ibrahim, N.F., Zainol, Z., Abdullah, R., Anbar, M., Alzubaidi, L., 2024. Evolving cybersecurity frontiers: A comprehensive survey on concept drift and feature dynamics aware machine and deep learning in intrusion detection systems. Eng. Appl. Artif. Intell. 137, 109143.

Skopik, F., Landauer, M., Wurzenberger, M., Vormayr, G., Milosevic, J., Fabini, J., Prüggler, W., Kruschitz, O., Widmann, B., Truckenthanner, K., et al., 2020. synERGY: Cross-correlation of operational and contextual data to timely detect and mitigate attacks to cyber-physical systems. J. Inf. Secur. Appl. 54, 102544.

Snehi, M., Bhandari, A., 2022. IoT-based DDoS on cyber physical systems: Research challenges, datasets and future prospects. In: International IOT, Electronics and Mechatronics Conference. pp. 1–8.

Strandberg, P.E., Söderman, D., Dehlaghi-Ghadim, A., Leon, M., Markovic, T., Punnekkat, S., Moghadam, M.H., Buffoni, D., 2023. The Westermo network traffic data set. Data Brief 50, 109512.

Talukder, M.A., Islam, M.M., Uddin, M.A., Hasan, K.F., Sharmin, S., Alyami, S.A., Moni, M.A., 2024. Machine learning-based network intrusion detection for big and imbalanced data using oversampling, stacking feature embedding and feature extraction. J. Big Data 11 (1), 33.

Tantawy, A., 2022. On the elements of datasets for cyber physical systems security. arXiv preprint arXiv:2208.08255.

Taormina, R., Galelli, S., Tippenhauer, N.O., Salomons, E., Ostfeld, A., Eliades, D.G., Aghashahi, M., Sundararajan, R., Pourahmadi, M., Banks, M.K., et al., 2018. Battle of the attack detection algorithms: Disclosing cyber attacks on water distribution networks. J. Water Resour. Plan. Manag. 144 (8), 04018048.

Termanini, A., Bourdoucen, H., Al-Abri, D., Al Maashri, A., 2026. Intrusion detection datasets for IIoT and ICS: A taxonomic review with a decision-aid scoring rubric. Sensors 26 (13), 4099.

Transportation Security Administration, 2018. Pipeline Security Guidelines. Transportation Security Administration, URL: https://www.tsa.gov/sites/default/files/pipeline_security_guidelines.pdf.

Tushkanova, O., Levshun, D., Branitskiy, A., Fedorchenko, E., Novikova, E., Kotenko, I., 2023. Detection of cyberattacks and anomalies in cyber-physical systems: Approaches, data sources, evaluation. Algorithms 16 (2), 85.

Umer, M., Sadiq, S., Karamti, H., Alhebshi, R.M., Alnowaiser, K., Eshmawi, A., Song, H., Ashraf, I., 2022. Deep learning-based intrusion detection methods in cyber-physical systems: Challenges and future trends. Electronics 11 (20), 3326.

Varshini, G.S., Latha, S., 2024. Detection and mitigation of coordinated cyber-physical attack in CPPS. Heliyon 10 (4).

Vigneshwaran, P., Thuseethan, S., Shanmugam, B., Thennadil, S., 2026. Cyber attack detection in smart grids: A survey of methods, challenges and future directions. Comput. Sci. Rev. 60, 100915.

Wang, Z., Xie, W., Wang, B., Tao, J., Wang, E., 2021. A survey on recent advanced research of CPS security. Appl. Sci. 11 (9), 3751.

Wollman, D.A., Weiss, M.A., Li-Baboud, Y., Griffor, E.R., Burns, M.J., 2017. Framework for Cyber-Physical Systems: Volume 3, Timing Annex. NIST Special Publication SP 1500-203, National Institute of Standards and Technology, Gaithersburg, MD, http://dx.doi.org/10.6028/NIST.SP.1500-203.

Wressnegger, C., Kellner, A., Rieck, K., 2018. Zoe: Content-based anomaly detection for industrial control systems. In: 48th International Conference on Dependable Systems and Networks. DSN, IEEE, pp. 127–138.

Xiao, S., Hu, J., Zhang, L., 2024. A multi-step attack path prediction method for oil & gas intelligence pipeline cyber physics system based on CPNE. Process. Saf. Environ. Prot. 185, 1303–1318.

Xu, L.D., Duan, L., 2019. Big data for cyber physical systems in industry 4.0: a survey. Enterp. Inf. Syst. 13 (2), 148–169.

Yang, Q., Hao, W., Ge, L., Ruan, W., Chi, F., 2019. FARIMA model-based communication traffic anomaly detection in intelligent electric power substations. IET Cyber Phys. Syst.: Theory Appl. 4 (1), 22–29.

Zahariadis, T., Papaestathiou, I., 2022. COSSIM - cyber-physical systems (CPS) simulator framework. Available at: https://omnetpp.org/download-items/COSSIM.html. (Accessed 12 October 2026).

Zemanek, S., Hacker, I., Wolsing, K., Wagner, E., Henze, M., Serror, M., 2022. PowerDuck: a GOOSE data set of cyberattacks in substations. In: 15th Workshop on Cyber Security Experimentation and Test. pp. 49–53.

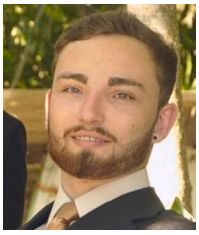
Zhang, Y., Liu, Q., 2022. On IoT intrusion detection based on data augmentation for enhancing learning on unbalanced samples. Future Gener. Comput. Syst. 133, 213–227.

Zhang, J., Pan, L., Han, Q.-L., Chen, C., Wen, S., Xiang, Y., 2021. Deep learning based attack detection for cyber-physical system cybersecurity: A survey. J. Autom. Sin. 9 (3), 377–391.

Zhou, X., Cheng, Z., Wang, C., Wang, S., Tao, C., Zhou, Z., Chen, X., Luo, J., Wang, D., Zhou, H., 2026. A dataset collected in real-world industrial control systems for network attack detection. Sci. Data 13 (1), 399.



Vagner E. Quincozes received his B.Sc. (2021) and M.Sc. (2023) degrees in Software Engineering from the Federal University of Pampa (UNIPAMPA). He is currently a Ph.D. student in Computing at Fluminense Federal University (UFF) and a Professor at Centro Universitário Serra dos Órgãos (UNIFESO). He has participated in research and development projects in partnership with DELL Brasil and UFF. His research interests include Artificial Intelligence, Machine Learning, Information Security, Software Engineering, Computer Networks, and the Internet of Things.



Silvio Ereno Quincozes received Software Engineer bachelors degree (2011) from the Federal University of Pampa (UNIPAMPA), a master's degree (2015) in Computer Science from the Federal University of Santa Maria (UFSM) and a Ph.D. student (from 2018) in Computing at the Fluminense Federal University (UFF). He is interested in researches topics related to Information Security, Computer Networks, Intrusion Detection Systems, Smart Grids, Digital Substation Systems, Internet of Things, Data Mining, and Software Engineering. He authored the best paper at the IX Latin American Network Operations and Management Symposium (LANOMS), in 2019.



Célio Albuquerque received the B.S. and M.S. degrees in electrical and electronics engineering from Universidade Federal do Rio de Janeiro, Brazil, in 1993 and 1995, and the M.S. and Ph.D. degrees in information and computer science from the University of California at Irvine in 1997 and 2000, respectively. From 2000 to 2003, he served as the networking architect for Magis Networks, designing high-speed wireless medium access control. Since 2004 he has been an Associate Professor at the Computer Science Department of Universidade Federal Fluminense, Brazil. His research interests include wireless networks, network security, Smart Grid communications, Internet architectures and protocols, multicast and multimedia services.



Diego Passos received his B.Sc., M.Sc. and D.Sc. degrees in Computer Science from Universidade Federal Fluminense (UFF), Rio de Janeiro, Brazil in 2007, 2009 and 2013, respectively. From 2013 to 2014, he worked as a postdoctoral fellow researcher at the same university. Later, he served as an associate professor at the Computer Science Department of UFF. Currently, he is a professor at Instituto Superior de Engenharia de Lisboa (ISEL). His research interests include multihop wireless networks, and wireless routing.



Dr. Mossé received his B.S. (Mathematics, 1985) from the University of Brasilia, Brazil, and M.S. and Ph.D. degrees (Computer Science, 1990 and 1993) from the University of Maryland, College Park. He has been a professor at the University of Pittsburgh since 1992, including six years as department chair, and has co-founded HiberSense, a startup company in the area of Smart Homes. He has been involved in the design and implementation of a couple of distributed, real-time operating system. His main research interest is in the allocation of resources (computing, energy, and network resources) in the realm of sustainable computing, computing for sustainability, and real-time, with main concerns being power management, security, and fault tolerance. He bridges the gap between the operating systems and networking research fields, between practice and theory. For the last 20 years, most of his systems research has focused on power and energy management, and for the last decade on how to increase diversity and how to promote reproducible research in computing.